Business cycle asymmetry of earnings pass-through

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Abstract

How does the insurance firms provide to their workers vary over the business cycle? Using Swedish administrative data, I document that the elasticity of earnings to positive idiosyncratic firm productivity shocks is acyclical, whereas that to negative ones doubles in recessions. A directed search model in which contracts balance insurance against retention reproduces both patterns: low job-finding rates in recessions lead firms to cut pay more after adverse shocks, while free entry caps raises after favorable ones in both states. The same forces generate procyclical skewness in earnings changes and imply that hiring subsidies, unlike unemployment benefits, erode workers' protection.

Keywords: Earnings pass-through, business cycles, directed search, income risk.
JEL codes: E24, E32, J31, J41, J64.

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1 Introduction

The extent to which firms insure their workers against salary fluctuations is a long-standing question in economics (Knight, 1921), and it rests on the idea that an unequal allocation of the match's surplus uncertainty between employers and employees is desirable if the two parties have different attitudes towards risk. For instance, a risk-neutral firm employing a risk-averse worker should optimally absorb fluctuations in the value of the match and pay stable compensation, thereby avoiding the risk premium that consumption volatility would command (Azariadis, 1975; Baily, 1974). Understanding whether and how much firms actually do so is thus key to gauging the income risk workers bear, and speaks directly to the frictions that might keep insurance from being complete and to the scope for public policy to intervene.

Starting with the seminal work by Guiso et al. (2005), and spurred in recent years by the increased availability of high-quality matched employer-employee data, several papers (for a review, see Guiso and Pistaferri, 2020) have tried to answer this question empirically, finding that firms shield workers' compensation well against the idiosyncratic productivity shocks they are exposed to, but also that the degree of insurance is not full, especially if the shocks are persistent. This literature has, however, predominantly studied the phenomenon unconditionally on the state of the economy. Insofar as the frictions that keep firms from fully insuring their workers bind differently depending on aggregate conditions, unconditional estimates would then miss a cyclical component of income risk.

This paper shows that this is indeed the case. Using Swedish registry data, I first document that the elasticity of earnings to positive idiosyncratic firm productivity shocks is invariant to the state of the economy, whereas that to negative ones roughly doubles in recessions. I then build a directed search model of the labour market that reproduces these patterns through two key features: the scarcity of outside job opportunities in recessions, which lets firms cut pay more after adverse shocks, and a ceiling on what competing employers can offer, which caps raises after favorable shocks regardless of aggregate conditions. Finally, I use the model to show that the same features generate a distribution of wage changes[1] with procyclical skewness and acyclical variance, and to study whether and how public policies targeting unemployment benefits and the cost of job creation can affect the insurance firms provide to their workers.

The empirical analysis relies on matched employer-employee data covering the universe of non-financial private companies, employment relationships, and workers in Sweden for the period 2004–2018. In my baseline, I measure idiosyncratic firm productivity shocks as innovations to value added per worker, and estimate their pass-through to the earnings of workers who remain at the same firm across two consecutive years. The average elasticity, 0.027, is in the range of the estimates reported in the literature and is stable across the cycle, but it blends two very different responses. Favorable shocks are passed through more than

[1] As shown below, labour supply in the model is fixed to unity, so there is no distinction between wages and earnings. I will therefore use the two terms interchangeably when referring to the model.

adverse ones on average, and at an elasticity that does not depend on aggregate conditions (0.032 in recessions against 0.039 in normal times, statistically indistinguishable), whereas adverse ones are largely absorbed by the firm in normal times (0.015) but passed through roughly twice as much in downturns (0.029). In monetary terms, a one standard deviation adverse shock reduces the annual salary of the average worker by about 2,900 SEK (in 2022 terms) in recessions, against 1,550 SEK in normal times. The asymmetry survives accounting for selection into staying,[2] and it is present among low-leverage firms, which are unlikely to be financially constrained, so that financial frictions cannot account for it on their own. It is also not uniform across workers: among the dimensions of heterogeneity I examine, tenure and earnings stand out, as in normal times the pass-through of adverse shocks declines with both, whereas in recessions the gradient reverses, and the loss of insurance falls on the workers who are costliest to dismiss.

To rationalize these patterns, I turn to a directed search model of the labour market with on-the-job search, risk-averse workers, and one-sided commitment, which follows Menzio and Shi (2010) in its structure and Balke and Lamadon (2022) in its focus on the transmission of firm shocks to wages. In each period, a matched firm must deliver the worker a given amount of utility, and optimally chooses how much to pay out today as a wage and how much to promise for tomorrow in each future state — promises that, since the worker is free to search on the job, also shape where she looks for alternative offers. The firm's choice thus entails a trade-off. On the one hand, insuring a risk-averse worker against wage fluctuations saves the volatility risk premium. On the other, not adjusting the terms of the contract means waiving the opportunity to align the worker's search incentives with the value of the match. To illustrate, consider a firm exposed to a persistent negative productivity shock. Since the latter reduces the expected value of the match, retaining the worker is worth less, so the firm optimally lowers the promise made for that state until the gain is balanced against the extra risk premium. Because the job-finding rate is decreasing in promised utility, the probability that the worker is poached rises, and because a lower promise is delivered through a lower wage, pay falls. For a firm exposed to a favorable shock the logic runs in reverse: retaining the worker is worth more, so the firm raises the promise to reduce the poaching risk, and the wage increases.

Two elements then generate the asymmetries found in the data. The first is a general equilibrium effect operating through market tightness, which accounts for the response to negative shocks. Because job-finding rates are depressed in downturns, inducing the same probability that the worker leaves requires a larger cut in promised utility, and hence in wages, than in normal times — the more so as lower aggregate productivity reduces the value of existing matches to begin with. The same mechanism would predict a larger upward response to favorable shocks in normal times, when competition for workers is stronger, a pattern the

[2] It is also robust to alternative definitions of business cycle episodes, shock extraction procedures, productivity measures, and controls for sectoral demand.

data do not show. What prevents it is the second element, free entry, which bounds the utility an entering firm can promise and still find the match worth forming. Since in the model only entrants can hire away a worker, an incumbent whose promise already stands at that bound faces no poaching risk at all. Firms hit by favorable shocks — which would want to raise promises to retain their workers — sit close to or above this threshold, so the retention motive drops out and only insurance remains, which caps their wage responses. Besides not binding on the downside, where firms are free to lower promises as far as the trade-off dictates, the cap holds in both aggregate states: in recessions the threshold falls with aggregate productivity, but most of the promises incumbents carry into the downturn were made in normal periods, where the economy spends most of its time, so firms hit by favorable shocks sit even more above it. Since the retention term is the only part of the firm's optimality condition that depends on the aggregate state, and it is inactive above the threshold, the pass-through of positive shocks is acyclical.

The data lend support to these mechanisms. As the retention motive strengthens with the shock, the model implies that the probability of switching employer falls the more favorable the shock. Such a pattern is indeed observed empirically, and is broadly replicated by the model even though only aggregate flow rates are targeted in the calibration. The match is closer for adverse shocks, however: mobility at large favorable ones is understated, because entrants are the only poachers and all start at the same productivity, so that workers at thriving firms cannot be lured away by more productive employers. This mismatch concerns levels rather than cyclicality: as in the data, switching displays no sizable difference between aggregate states anywhere in the shock distribution.

The mechanisms just described have implications beyond matching the empirically documented pass-through asymmetries. First, the model provides a new explanation for the business cycle trends in income risk documented by a recent empirical literature (Busch et al., 2022; Guvenen et al., 2014). Since adverse shocks are passed through more in recessions while the transmission of favorable ones is acyclical, the left tail of the wage growth distribution of stayers thickens in downturns and thins in the recovery. Dispersion, however, barely moves: cuts become more frequent but not deeper, and raises become rarer, so the mass that shifts stays close to the mean, where it contributes little to the variance. The model-generated distribution thus features procyclical skewness and acyclical variance, as in the data.

Second, the model speaks to how policy can, and cannot, shape the insurance firms privately provide, and thus the part of income risk that originates from the employer. Varying the generosity of unemployment benefits leaves pass-through and its cyclical asymmetry essentially unchanged: a more generous benefit shifts the level of match values and of the promises over which the parties contract, but not the marginal condition that pins down wages, and it leaves the job-to-job rate — and hence the poaching threat existing matches face — unchanged. This result thus offers a rationale for the evidence of Jäger et al. (2020) that

incumbents' wages respond little to the value of nonemployment. The cost of job creation, by contrast, acts on the poaching threat directly, a channel that adds to its effect through the reservation wage of the unemployed identified by Cahuc et al. (2019). When hiring is expensive, few vacancies are posted and workers are rarely poached in either aggregate state: a recession then makes little further difference to the threat incumbents face, and the cyclical gap closes. A higher cost of job creation thus involves a trade-off: it makes firms better insurers of incumbent workers, and especially so in recessions, when that protection matters most, but at the expense of lower job-finding and job-to-job mobility.

Relation to the literature. First, this paper contributes to the empirical research on the pass-through of idiosyncratic firm shocks to workers' compensation, whose main finding — which my estimates confirm — is that stayers are overall well insured against them (for a review, see Guiso and Pistaferri, 2020). Recent work has addressed the selection into staying that these estimates are subject to (Chan et al., Forthcoming; Friedrich et al., Forthcoming), documented heterogeneity in pass-through by worker and firm characteristics, including firms' financial constraints (Chan et al., Forthcoming; Friedrich et al., Forthcoming; Juhn et al., 2018; Malgieri and Citino, 2025), and linked the tails of the earnings growth distribution to firm outcomes, showing that large losses become more frequent in shrinking firms also among stayers (Grübener and Rozsypal, 2026). A separate body of work studies the transmission of sectoral and aggregate shocks. Carlsson et al. (2016) and Souchier (2023) find, in Swedish and French data respectively, that sectoral shocks are passed through considerably more than firm-level ones, and Braggion et al. (2025) estimate the pass-through of aggregate fluctuations themselves, finding that it is lower for high-income workers and permanent contracts, operates mainly through hours, and is smaller in recessions than in expansions. My contribution lies at the intersection of these two literatures: I study how the transmission of idiosyncratic shocks depends on the aggregate state, separately for favorable and adverse shocks, and I document how this dependence varies across worker and firm characteristics. The closest paper to my analysis is Chan et al. (Forthcoming), who report, using Danish wage data, that persistent negative firm shocks are always passed through and positive ones only outside recessions. The difference with my findings plausibly reflects the compensation measure. Collective agreements constrain the negotiated wage, especially in downturns, whereas hours and variable pay arguably remain more adjustable in either direction. In recessions, then, both cuts after adverse shocks and raises after favorable ones go through hours and bonuses, which show up in earnings but not in wages.[3]

[3]In subsequent work, Meeuwis et al. (2026) find in US data that the pass-through of idiosyncratic firm shocks rises with aggregate risk premia. The amplification they document operates mainly through job loss among lower-paid workers rather than through the pay of stayers, consistent with US firms adjusting on the extensive margin, unlike Swedish ones, for which stricter employment protection makes that margin costlier and leaves more of the adjustment to workers' compensation.

Second, this study relates to the theoretical literature on long-term employment contracts. Building on the seminal contributions of Baily (1974) and Azariadis (1975), Thomas and Worrall (1988) derived contracts that must be self-enforcing under limited commitment. The present paper belongs to the strand that applies these contracts in directed search economies: Rudanko (2009) first did so in a model with aggregate shocks, finding that the resulting wage rigidity is consistent with aggregate wage data only when neither party can commit; Menzio and Shi (2010, 2011) showed that directed search makes the equilibrium of these economies block recursive, so that the latter can be solved without tracking the distribution of workers across employment states; and Schaal (2017) used this class of models to study the effects of firm-level uncertainty. Within this strand, it builds in particular on Balke and Lamadon (2022), who rationalize the pass-through of firm shocks through the firm's trade-off between insuring workers and shaping their on-the-job search incentives.

Several recent papers study this mechanism in economies with aggregate fluctuations, each emphasizing a different driver. In particular, contemporaneous work by Souchier (2023) shares the key equilibrium force with this paper, the competition for workers. In his framework, favorable sectoral shocks intensify that competition, so firms must raise wages more to retain workers, whereas firm-level shocks leave it unchanged and are passed through much less. I show that, because the same competition varies with the aggregate state, the interaction between aggregate and firm-level shocks alters the transmission of the latter, and that this mechanism, together with the bound free entry places on firms' promises, accounts for the asymmetries I document empirically. Meeuwis et al. (2026) also use a similar framework to show that higher risk premia generate a larger pass-through of negative idiosyncratic firm shocks; Acabbi et al. (2026) add two-sided heterogeneity and firm-dependent human capital accumulation; and Malgieri and Citino (2025) tie wage dynamics to the firm's financial constraints rather than to on-the-job search. Besides focusing on the business cycle, my paper separately accounts for the response to favorable shocks, which none of these works examines.

Third, this work relates to the literature documenting how income risk varies over the business cycle, which finds procyclical skewness and acyclical variance in the distribution of income shocks in US administrative data (Guvenen et al., 2014) as well as in Sweden, Germany and France, including among continuously employed full-time workers (Busch et al., 2022). Existing accounts of the cyclicality of skewness attribute it to margins outside the ongoing employment relationship, such as the earnings changes of employer and occupation switchers (Carrillo-Tudela et al., 2022). This paper shows instead that these patterns arise even among workers who do not change employer, from the way firms transmit their own shocks over the cycle.

Finally, the policy analysis relates to the study of how public intervention interacts with the insurance firms privately provide, which Balke and Lamadon (2022) examine for progressive taxation in an economy without aggregate fluctuations. I extend this question to an economy

with business cycles, studying unemployment benefits and the cost of job creation, and show that only the latter changes how firms transmit their shocks to wages, especially in recessions.

Structure of the paper. Section 2 presents the empirical findings. Section 3 describes the model and Section 4 its calibration and estimation. Section 5 inspects the model's mechanisms and assesses its fit to the pass-through patterns. Section 6 studies the implications for cyclical income risk and for the interaction between public policies and firm-provided insurance. Section 7 concludes.

2 Empirical analysis

The empirical part of this paper is based on a matched employer-employee dataset built by merging three administrative registers assembled by Statistics Sweden and accessed through the Institute for Evaluation of Labour Market and Education Policy (*Institutet för arbetsmarknads- och utbildningspolitisk utvärdering*, IFAU).

The first is the Structural Business Statistics (*Företagens ekonomi*, FEK), which contains accounting information for the universe of non-financial corporations in Sweden starting from 1997. From this dataset I recover, in addition to balance sheet items, a value-added measure constructed by Statistics Sweden and each firm's operating sector.

The second source is the Register-Based Labour Market Statistics (*Registerbaserad arbetsmarknadsstatistik*, RAMS), which covers all employment relationships in Sweden from 1985 onward. From this register I collect firms' legal form and municipality and, on the workers' side, salaries, the duration of employment relationships, and occupational status.

The last source is the Longitudinal Integrated Database for Health Insurance and Labour Market Studies (*Longitudinell integrationsdatabas för sjukförsäkrings- och arbetsmarknadsstudier*, LISA), which provides socio-economic and demographic information on the Swedish population from 1990 onward, and from which I recover civil status, gender, year of birth, and education. Appendix B provides further details on the data sources and on the construction of the final dataset.

2.1 Sample description

The period of the analysis is 2004–2018. I classify an individual as employed at a firm if they work there for at least six months in a year and, for those with multiple employers, assign as their unique workplace in each year the firm paying the highest salary. Earnings are annualized when the worker is at the firm for less than the full year, and firm-level employment is the number of workers classified as employed at the firm. I restrict the sample to limited liability companies operating in manufacturing, construction, trade, and services with at least five employees and positive value added, and to individuals aged 25 to 55 whose occupational

status is employee.[4] Firm balance sheet variables are deflated with the GDP deflator and salaries with the CPI, both with 2022 as the reference year.

As shown in Table 1, the final sample includes about 3.3 million unique workers and 160 thousand unique firms, for a total of more than 20 million worker-year and 1 million firm-year observations. On average, it includes about 68 thousand firms and 1.46 million workers per year, covering about 65 percent of value added and 81 percent of employment in the sectors considered.[5] The table also reports that the average worker is about forty years old, has 6.5 years of tenure, and earns roughly 422 thousand SEK per year, and that the vast majority of workers are stayers, i.e., they remain employed at the same firm across two consecutive years.[6] Average value added per worker is about 692 thousand SEK, with an average annual growth rate of 2.5 percent. Both earnings and value added per worker overall increase with firm size and are broadly comparable across sectors.

[4]The matched data potentially go back to 1997, but the introduction of standardized accounting records as a source of information in 2004 caused a change in the total number of firms included. To avoid potential inconsistencies due to this change, I start the analysis in 2004. In addition to the criteria above, I exclude observations with implausible or extreme balance-sheet or earnings values. The full list of restrictions is in Appendix B.

[5]When benchmarked against the whole non-financial private sector and the whole economy, coverage is, respectively, about 60 and 45 percent for value added and 77 and 49 percent for employment. Appendix Figure A.1 shows the evolution of the number of firms and workers in the sample by broad sector and firm-size group.

[6]The table reports the share of stayers including exiters. When restricting to workers observed in the sample in the next period (stayers or switchers), the share is 0.89.

Firm-level average productivity and worker compensation by sector and size					
	Firm size				
	5–19	*20–49*	*50–99*	*100+*	*All*
Manufacturing					
VA/worker	653.3	742.0	894.4	1078.9	723.8
VA/worker, log growth	0.024	0.004	0.000	-0.004	0.015
Worker earnings	374.2	396.9	419.1	474.8	449.6
Construction					
VA/worker	631.1	701.7	771.3	818.3	649.4
VA/worker, log growth	0.040	0.013	0.003	-0.007	0.033
Worker earnings	382.2	419.9	453.6	494.2	436.4
Trade					
VA/worker	632.5	776.2	816.5	896.3	669.7
VA/worker, log growth	0.022	-0.002	-0.001	-0.005	0.016
Worker earnings	373.6	426.5	434.2	406.9	404.5
Services					
VA/worker	694.1	713.0	766.0	831.6	707.5
VA/worker, log growth	0.045	0.000	0.003	-0.006	0.031
Worker earnings	385.9	416.1	426.4	414.6	410.5
All sectors					
VA/worker	661.7	730.5	807.4	914.1	691.6
VA/worker, log growth	0.035	0.002	0.001	-0.005	0.025
Worker earnings	380.4	415.0	428.6	438.2	422.3

Worker characteristics		**Sample size**	
Average worker age	39.7	Unique workers	3,297,393
Share edu > high school	0.35	Unique firms	161,530
Share married	0.42	Worker-year obs.	21,894,317
Average worker tenure (years)	6.5	Firm-year obs.	1,021,369
Share female	0.35		
Fraction stayers	0.77		

Table 1: Summary statistics. Sample period 2004–2018. Monetary values in thousands of 2022 SEK.

2.2 Measuring earnings pass-through over the cycle

The aim of the empirical analysis is to examine whether idiosyncratic firm productivity shocks are passed through workers' earnings differently over the business cycle.[7] This requires three ingredients: a classification of years into recessionary and non-recessionary, a measure of idiosyncratic firm productivity shocks, and an econometric specification relating earnings changes to these shocks. Each is described in turn below.

Business cycle episodes. Figure 1 plots real GDP growth and the unemployment rate in Sweden over the sample period. GDP growth was negative in only three years, 2008, 2009, and 2012. Unemployment rose in the aftermath of the Great Recession, declined thereafter,

[7] A precondition for firm-level pass-through is that pay is not fully set by central agreements. As documented in Appendix W.3.3 of Balke and Lamadon (2022), a substantial share of Swedish employment contracts involves a firm-level bargaining component, so that even negotiated wages can reflect firm performance.

and rose again slightly during the European debt crisis. From 2016 to the end of the sample, growth slowed but remained positive, while unemployment continued to decline.

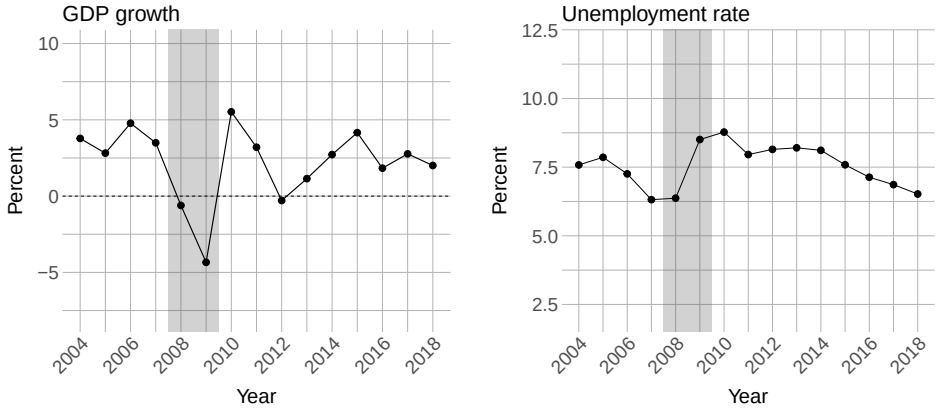


Figure 1: Aggregate macroeconomic conditions. Real GDP growth (left) and unemployment rate (right) in Sweden, 2004–2018. Shaded areas denote the years classified as recessions in the baseline analysis (2008–2009).

Based on these patterns, I classify 2008-2009 as recessionary years (R) and the remaining ones as non-recessionary (NR). Section 2.3.1 shows that the results are robust to alternative definitions of business cycle phases.

Idiosyncratic firm productivity shocks. Let $\tilde{z}_{jt}$ denote the productivity of firm j in year t, measured in the baseline as the log of value added per worker. Following the approach of Bloom et al. (2018),[8] I identify idiosyncratic firm productivity shocks as the residual ν_{jt} of

$$\tilde{z}_{jt} = \rho \tilde{z}_{jt-1} + \mu_j + \lambda_{s(j)t} + \nu_{jt}, \tag{1}$$

where μ_j and $\lambda_{s(j)t}$ are firm and 2-digit sector-by-year fixed effects, respectively. I exclude sector-year cells with fewer than ten firms and, in the baseline, retain firms with at least five usable observations.[9]

Pass-through regressions. Let w_{ijt} indicate the annual earnings of worker i employed at firm j in year t, and $\Delta \log w_{ijt}$ the change in its log between two consecutive periods. To estimate the pass-through of idiosyncratic firm productivity shocks to workers' earnings, I run the regression

$$\Delta \log w_{ijt} = \beta_0 + \beta_1 \nu_{jt} + \beta_2 \nu_{jt} \cdot \mathbb{1}\{\nu_{jt} < 0\} + \beta_3 X_{it} + \beta_4 Z_{jt} + \lambda_t + \varepsilon_{ijt}, \tag{2}$$

where ν_{jt} is the firm shock from equation (1), λ_t are year fixed effects, and X_{it} and Z_{jt} collect worker and firm characteristics. Worker controls include third-order polynomials in lagged

[8] This approach recovers a realized shock for each firm-year, which is what allows estimating pass-through separately by the sign of the shock. The moment-based estimator of Guiso et al. (2005) instead identifies the pass-through of permanent and transitory shocks from covariances of earnings and value-added changes which, as the authors note, rules out asymmetric responses, since with both components present the sign of each is not observed.

[9] As Bloom et al. (2018) acknowledge, this revenue-based measure may capture not only pure productivity but also demand shocks. The robustness analysis of Section 2.3.1 investigates in more detail this and other aspects related to measuring productivity.

age and lagged tenure, a gender dummy, and dummies interacting age group with education level, while firm controls include lagged firm productivity and dummies for region (*län*), firm size, and broad sector.[10] The regression is estimated on stayers, i.e., workers employed at the same firm in two consecutive years, and, to limit the impact of outliers, excludes observations with earnings changes or firm shocks below the 0.5th or above the 99.5th percentile of their respective distributions within each year.

2.3 Results

Figure 2 gives a first, descriptive look at the main patterns. The left panel plots average earnings changes by bin of the firm's idiosyncratic productivity shock, separately for recessionary and non-recessionary years. For positive shocks, the two lines are nearly indistinguishable: earnings gains rise with firm performance, and the slope does not depend on the state of the cycle. For negative ones, instead, the lines diverge: earnings changes decline more steeply as shocks worsen in recessions, whereas in non-recessionary years the relationship flattens, suggesting that workers are better insulated from adverse shocks in normal times. Consistent with this interpretation, and despite average earnings changes remaining positive in virtually all shock bins,[11] the right panel shows that the share of workers experiencing a negative earnings change, between roughly 30 and 40 percent, rises as firm shocks deteriorate in both phases, but more markedly in recessions.

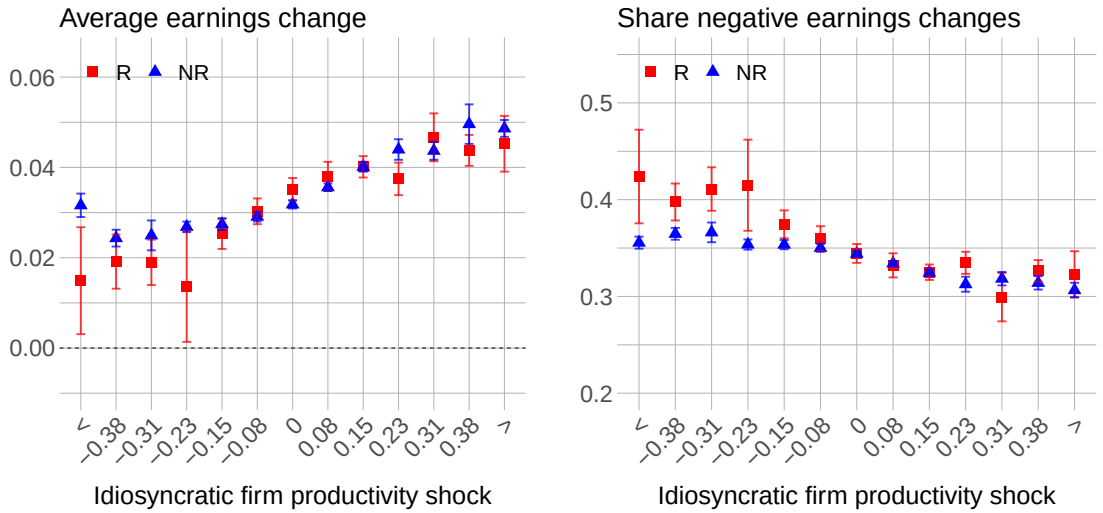


Figure 2: Earnings changes and firm shocks. Average log earnings change of stayers (left) and share of stayers with a negative earnings change (right), by bin of the firm's idiosyncratic productivity shock, in recessions (squares) and non-recessions (triangles). Vertical bars denote 90% confidence intervals based on standard errors clustered at the firm level. Firm shocks are trimmed at the 0.5th and 99.5th percentiles within each year in both panels, and earnings changes in the left panel only. Edge bins collect shocks beyond the outermost interior bins.

[10] Age groups are five-year bins from 25 to 54, with the last bin also including age 55; education levels are pre-secondary (9 years or less), secondary, post-secondary shorter than two years, and post-secondary of two years or longer (including postgraduate); firm size classes are 5–19, 20–49, 50–99, and 100+ employees; broad sectors are manufacturing, construction, trade, and services.

[11] This could reflect, for instance, returns to experience and tenure, as well as aggregate real wage growth.

Table 2 formalizes this evidence. Columns 1–3 report estimates from a symmetric specification, in which positive and negative shocks are constrained to have the same effect. Pooling all years, the pass-through elasticity is positive and equal to 0.027, within the range of estimates reported in the literature (Guiso and Pistaferri, 2020). It is also stable across the cycle: the recession and non-recession estimates, 0.030 and 0.026, are not statistically distinguishable (p-value 0.378). Relaxing the symmetry restriction (columns 4–6), though, reveals that this average blends two very different responses: the pass-through of positive shocks (0.038) is roughly twice that of negative ones (0.019), which shows that, on average, workers are substantially insulated from adverse shocks to their employer.

This asymmetry, however, varies over the cycle. In non-recessionary years the gap between positive and negative pass-through is large and significant (−0.024), whereas in recessions it essentially disappears (−0.003, insignificant). This reflects a change on the negative side only: the pass-through of positive shocks is similar in the two states (0.039 versus 0.032, p-value 0.149), while that of negative ones nearly doubles in downturns (0.015 versus 0.029, p-value 0.091). In monetary terms, these elasticities imply that a one-standard-deviation adverse shock reduces annual earnings by roughly 2,900 SEK in recessions, against 1,550 SEK in non-recessionary years. In sum, the protection that workers enjoy against adverse firm shocks in normal periods erodes in bad times, while the transmission of favorable ones is unchanged.

Dependent variable: $\Delta \log w_{ijt}$, log earnings change						
	Average Effect			Asymmetric Effects by Shock Sign		
	(1) All Years	(2) Recession	(3) Non-recession	(4) All Years	(5) Recession	(6) Non-recession
ν_{jt} (β_1)	0.027***	0.030***	0.026***	0.038***	0.032***	0.039***
	(0.002)	(0.004)	(0.001)	(0.002)	(0.005)	(0.002)
$\nu_{jt} \cdot \mathbb{1}\{\nu_{jt} < 0\}$ (β_2)				-0.019***	-0.003	-0.024***
				(0.003)	(0.011)	(0.003)
Sum: $\beta_1 + \beta_2$				0.019***	0.029***	0.015***
				(0.003)	(0.008)	(0.002)
R^2	0.008	0.010	0.008	0.008	0.010	0.008
Obs. (Millions)	14.3	2.0	12.3	14.3	2.0	12.3
Monetary effects (SEK)						
β_1	2785	3081	2659	3893	3294	3993
$\beta_1 + \beta_2$				1911	2946	1553
Equality Tests (p-value)						
$H_0 : \beta_1^R = \beta_1^{NR}$		0.378			0.149	
$H_0 : (\beta_1 + \beta_2)^R = (\beta_1 + \beta_2)^{NR}$					0.091	

Table 2: Pass-through of firm shocks to earnings. Estimates of equation (2). Columns 1–3 impose a symmetric response and columns 4–6 allow it to differ by the shock's sign; within each block, the first column pools all years and the second and third use recession and non-recession years, respectively. Standard errors clustered at the firm level in parentheses; $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. The equality tests report p-values for the difference between recession and non-recession coefficients, obtained from a pooled specification interacting the regressors with the business cycle state. Monetary equivalents, expressed in 2022 SEK, multiply the relevant coefficient by one standard deviation of the shock and by average earnings in the corresponding subsample.

2.3.1 Robustness

I now examine the robustness of the baseline results to sample selection, the definition of business cycle episodes, the procedure used to extract idiosyncratic firm shocks, the measurement of firm productivity, and sectoral demand, and I assess whether financial frictions could be an alternative explanation for the cyclical asymmetry. Sample selection and financial frictions are discussed below; the remaining checks are reported in Appendix C. Overall, the baseline results are robust throughout.

Sample selection. As the estimation sample consists of stayers, a natural question is whether selection biases the estimates.[12] For instance, after an adverse shock workers are more likely to leave — to another employer, especially in non-recessions when jobs are plentiful, or to unemployment, especially in recessions — and if the leavers are the ones facing the largest earnings losses, the pass-through estimated on stayers is attenuated. Likewise, firms receiving favorable shocks may find it harder to retain workers in non-recessions, when competition for them is stronger, and if the leavers are those who would have gained the most, the pass-through is again attenuated.

Figure 3 provides a first look at the issue by plotting the fraction of workers leaving the firm in the year of the shock, across the two business cycle states. Leavers include both those who switch to another employer and those who exit the sample (because they become unemployed, migrate, or move to employers outside my sample restrictions).[13] The probability of leaving is mildly decreasing in the shock throughout, except at large positive shocks, where it slightly rises. There is, however, no evident gap between the two business cycle states. Restricting to job-to-job switchers within the sample (Appendix Figure A.2) reveals a similar pattern. Selection is thus present, particularly at large shocks, but does not differ across business cycle states, and is therefore unlikely to drive the cyclical asymmetry.

[12] This concern is well known, and two recent papers address it directly: Friedrich et al. (Forthcoming) model workers' mobility decisions explicitly, and Chan et al. (Forthcoming) correct for selection using family linkages.

[13] As the analysis conditions on the firm's current shock, selection through firm exit falls outside this framework. Firm-side attrition is, however, nearly acyclical: the share of firms present in the regression sample in one year but not in the next is about 13% overall and roughly 14% in recessions.

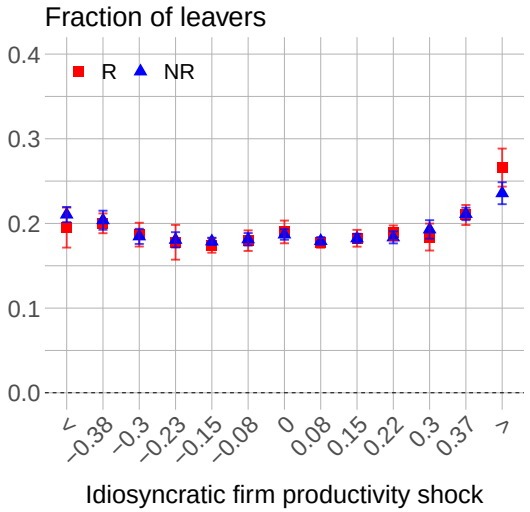


Figure 3: Job leaving and firm shocks. Fraction of workers who either change employer or exit the sample by bin of the origin firm's idiosyncratic productivity shock in the transition year, in recessions (squares) and non-recessions (triangles). Vertical bars denote 90% confidence intervals based on standard errors clustered at the firm level. Shocks trimmed at the 0.5th and 99.5th percentiles within each year. Edge bins collect shocks beyond the outermost interior bins.

To test this formally, I follow Chan et al. (Forthcoming) and control for selection using civil status as instrument for the staying decision. The identifying assumption is that civil status and its changes affect the decision to remain at the current firm but, conditional on staying, not the response of earnings to firm shocks. I estimate a probit model for the probability of staying,

$$\Pr(S_{ijt} = 1) = \Phi(\gamma_0 + \gamma_1 \nu_{jt} + \gamma_2 \nu_{jt} \cdot \mathbb{1}\{\nu_{jt} < 0\} + \gamma_3 X_{it} + \gamma_4 Z_{jt} + \gamma_5 M_{it} + \lambda_t), \qquad (3)$$

where $S_{ijt} = 1$ if worker i stays at firm j between $t-1$ and t, X_{it} and Z_{jt} contain the same variables and dummies as those included in equation (2) (all measured at $t-1$), and M_{it} includes dummy variables indicating whether an individual at time t is married, separated, or changed spouse. I estimate the probit separately for recessionary and non-recessionary years, recover the inverse Mills ratio (IMR), and add it to equation (2) together with its interaction with the negative-shock indicator, which allows the correction to differ by the shock's sign, in line with the mobility patterns of Figure 3.

Dependent variable: $\Delta \log w_{ijt}$, log earnings change		
	Recession	Non-recession
β_1 ($\nu_{jt} \geq 0$)	0.011*	0.011***
	(0.006)	(0.003)
$\beta_1 + \beta_2$ ($\nu_{jt} < 0$)	0.035***	0.019***
	(0.009)	(0.003)
IMR	0.158***	0.235***
	(0.037)	(0.017)
IMR $\times \mathbb{1}\{\nu_{jt} < 0\}$	−0.008	−0.021***
	(0.010)	(0.003)
R^2	0.010	0.008
Obs. (Millions)	2.0	12.3
Equality Tests (p-values)		
$H_0 : \beta_1^R = \beta_1^{NR}$	0.953	
$H_0 : (\beta_1 + \beta_2)^R = (\beta_1 + \beta_2)^{NR}$	0.090	

Table 3: Pass-through corrected for selection into staying. Estimates of equation (2) with asymmetric effects, by recession and non-recession years, augmented with the inverse Mills ratio (IMR) from the probit model of equation (3) and its interaction with the negative-shock indicator. Standard errors clustered at the firm level in parentheses; $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Equality tests report p-values for the recession–non-recession difference from a pooled specification interacting the regressors with the business cycle state.

Table 3 reports the corrected estimates. Three findings emerge. First, consistent with Figure 3, in my sample the IMR coefficient is positive and significant in both states, indicating positive selection into staying: workers who remain despite a low predicted probability of staying have systematically higher earnings growth. The negative coefficient on its interaction with the negative-shock indicator, shows a selection effect for adverse shocks smaller than for favorable ones. Second, the correction matters mostly on the positive side, where mobility rises sharply at large shocks: the pass-through falls to 0.011 in both states. It remains positive and significant, however, and identical across states, so the acyclicality of the positive-side response is unaffected. Third, the negative-shock coefficients rise only marginally, and the cyclical asymmetry is unchanged. Selection thus slightly affects the level of pass-through, particularly for favorable shocks, but not its cyclical properties.

Financial frictions. Financial frictions offer an alternative explanation for the cyclical asymmetry: constrained firms may find it harder to absorb adverse shocks in downturns, when access to external finance tightens, and may thus be forced to pass more of them through to workers. Figure 4 assesses this possibility by splitting firms by quintile of pre-shock leverage. The recession pass-through of adverse shocks exceeds its non-recession counterpart especially at high-leverage firms, consistent with a role for financial frictions. The gap does not vanish at the low end of the leverage distribution, however, where firms are unlikely to be constrained, so this channel may contribute to the pattern but does not appear to account for it on its own.[14] Consistently, the recession increase in the pass-through of adverse shocks is also most

[14]For favorable shocks, by contrast, there is no gap between recessions and non-recessions across leverage quintiles (Appendix Figure A.3).

pronounced at large firms (Appendix Figure D.6), where constraints are least likely to bind.

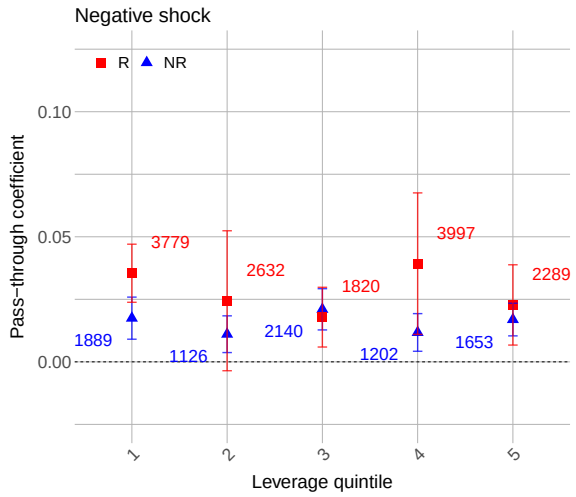


Figure 4: Pass-through by firm leverage. Pass-through coefficients for negative firm shocks, estimated separately by quintile of lagged firm leverage (quintile 1 = lowest leverage), in recessions and non-recessions. Vertical bars denote 90% confidence intervals based on standard errors clustered at the firm level. Monetary values are expressed in 2022 SEK and computed as the coefficient times the standard deviation of the firm shock times average earnings in each group.

2.3.2 Who loses insurance in recessions?

The cyclical erosion of insurance documented above is not uniform across workers, as Figure 5 illustrates by reporting the pass-through of negative shocks by state for groups defined by lagged tenure (left panel) and lagged earnings quintile (right panel). In non-recessionary years the elasticity declines in both variables — workers with the longest tenure and the highest earnings face the smallest pass-through of adverse shocks — while in recessions the gradient reverses, with pass-through rising in both. This suggests that the loss of insurance in downturns falls on the workers who are costliest to dismiss. The transmission of positive shocks, by contrast, shows no gap between business cycle states across groups and is broadly flat in both variables, apart from a larger elasticity in the lowest earnings quintile (Appendix Figure D.6). Workers with longer tenure and higher earnings thus enjoy slightly greater protection from losses in normal times without capturing a larger share of gains. Patterns across further worker and firm characteristics — age, education, sector, firm size, and firm productivity — are reported in Appendix D.

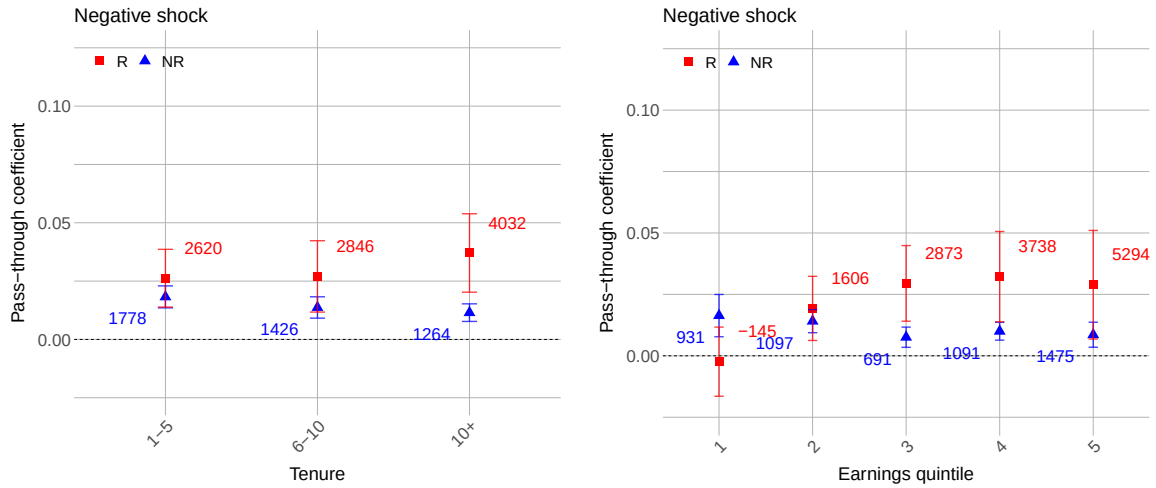


Figure 5: Who loses insurance in recessions? Pass-through of negative firm shocks by lagged tenure group (left) and lagged earnings quintile (right), in recessions (squares) and non-recessions (triangles). Vertical bars denote 90% confidence intervals based on standard errors clustered at the firm level.

The next section develops a directed search model with on-the-job search that rationalizes both the acyclicality of the response to positive shocks and the larger transmission of negative ones in recessions.

3 Model

This section develops a directed search model of the labour market in which to study the transmission of idiosyncratic firm shocks to workers' wages. The environment follows Menzio and Shi (2010) closely, and its focus on the labour market consequences of firm shocks is close in spirit to Balke and Lamadon (2022).

Environment. The model economy is inhabited by a continuum of infinitely lived ex-ante identical workers and a positive measure of firms. Firms maximize the present discounted value of profits at rate $\beta \in (0,1)$ by transforming with a constant returns to scale technology one unit of labour into $f(y,z)$ units of output, where y denotes aggregate and z idiosyncratic productivity. Both take values on finite grids, $y \in Y = \{y_1, \ldots, y_{N_y}\}$ with N_y finite and $\underline{y} = y_1 < \cdots < y_{N_y} = \overline{y}$ and $z \in Z = \{z_1, \ldots, z_{N_z}\}$ with N_z finite and $\underline{z} = z_1 < \cdots < z_{N_z} = \overline{z}$. Workers derive utility from consumption according to the function $u(\cdot)$ - with $u : \mathbb{R} \to \mathbb{R}$ twice continuously differentiable, strictly increasing and weakly concave function - and maximize the present discounted value of utility at rate β. A worker is either employed, in which case she consumes the wage paid by her firm, or unemployed, in which case she consumes the unemployment benefit b. There is no saving technology.

The aggregate state of nature s follows a finite-state Markov chain with transition probabilities $\Pi_s(\hat{s}|s)$, where hats denote next-period values, and determines aggregate productivity $y(s)$. Idiosyncratic productivity z likewise follows a finite-state Markov chain, with transition

probabilities $\Pi_z(\hat{z}|z)$. The aggregate state of this economy ψ, includes the aggregate state of nature s but also the distribution of workers across unemployment and employment states, respectively $g_u \in [0,1]$ and $g_e : X \times Z \to [0,1]$ with $g_e(V,z)$ being the share of workers employed at a firm with idiosyncratic productivity z under a contract that guarantees to the worker an expected lifetime utility of V.

Workers search for jobs and firms post vacancies in a continuum of submarkets indexed by $x \in X = \left[\underline{x}, \overline{x}\right]$, the expected utility the firm promises a worker hired in that submarket, with $\underline{x} < u(b)/(1-\beta)$ and $\overline{x} > u(f(\overline{y},\overline{z}))/(1-\beta)$. Market tightness — the ratio of vacancies to workers searching for a job — in each submarket depends on the aggregate state and is denoted by $\theta(\psi, x)$.

Each period has four stages: separation, search, matching, and production. In the separation stage, existing matches are destroyed exogenously with probability $\delta \in (0,1)$. In the search stage, firms choose how many vacancies to post and in which submarket, at per-vacancy cost $k > 0$, while workers choose in which submarket to search, with intensities $\lambda_u, \lambda_e \in (0,1]$ when unemployed and employed, respectively. In the matching stage, workers and firms in each submarket are matched according to a constant returns to scale function: in a submarket with tightness θ, a worker finds a job with probability $p(\theta)$ and a vacancy is filled with probability $q(\theta) = p(\theta)/\theta$. The function $p : \mathbb{R}_+ \to [0,1]$ is twice continuously differentiable, strictly increasing, strictly concave, with $p(0) = 0$ and $p'(0) < \infty$, and $q : \mathbb{R}_+ \to [0,1]$ is twice continuously differentiable, strictly decreasing, convex function, with $q(0) = 1$ and $q'(0) < 0$. All new matches start at the same idiosyncratic productivity $z_0 \in Z$. In the production stage, an employed worker produces $f(y,z)$ units of output and consumes the wage w specified by the employment contract, while unemployed workers consume the benefit $b \in (0, f(\overline{y},\overline{z}))$. Finally, before the next period, the new aggregate state $\hat{s}$ and the new idiosyncratic productivity $\hat{z}$ are drawn.

Contracts. Upon matching, the relationship is governed by a contract, chosen by the firm to maximize profits, that specifies a series of wages $\{w_{t+j}\}_{j=0}^{\infty}$ for every possible history (s^{t+j}, z^{t+j}). Following Spear and Srivastava (1987), the firm's problem can be written recursively by adding the promised utility V as a state variable: in each period, the firm chooses the current wage w and the expected lifetime utility $\hat{V}(\hat{\psi}, \hat{z})$ it promises the worker in each future state. As explained below, the worker's optimal on-the-job-search depends on the utility she expects from her current relationship, so $\hat{V}(\hat{\psi}, \hat{z})$ also governs her search once that state is realized. Commitment is one-sided: the firm must deliver the promised utility, though it is free to choose the split between the current wage and future promises, while the worker is free to search on the job. The firm therefore chooses $\hat{V}(\hat{\psi}, \hat{z})$ taking into account the search behavior it induces on the worker's side.

Worker's problem. An employed worker with promised expected lifetime utility V who has the opportunity to search on the job must choose a submarket x. With probability $p(\theta(\psi, x))$ she finds a new job with expected lifetime utility x and otherwise remains in her current job, receiving the utility specified by the contract. Her lifetime utility at the beginning of the search stage is thus $V + \max\{0, R(\psi, V)\}$ where R is the return to search,

$$R(\psi, V) = \max_{x \in X} p\left(\theta(\psi, x)\right)(x - V). \tag{4}$$

The solution yields the optimal search choice $x^*(\psi, V)$ and the implied probability of leaving $\tilde{p}(\psi, V) := p\left(\theta(\psi, x^*(\psi, V))\right)$, where x^* balances the directed-search trade-off between promised utility and the probability of obtaining it — higher promises are offered in submarkets with lower job-finding probability.

An unemployed worker, instead, consumes the benefit b in the production stage and, in the next period, searches with probability λ_u. Her value is, therefore:

$$U(\psi) = u(b) + \beta \mathbb{E}\left[U(\hat{\psi}) + \lambda_u \max\{0, R(\hat{\psi}, U(\hat{\psi}))\}\right]. \tag{5}$$

Firm's problem. A firm matched with a worker to whom it has promised lifetime utility V, when the aggregate state of nature is s and idiosyncratic productivity is z, chooses the current wage w and the promised utility $\hat{V}$ in each future state to maximize profits, subject to delivering V. Letting $\xi = (\psi, z)$, the firm's value is

$$J(\xi, V) = \max_{w, \{\hat{V}(\hat{\xi})\}_{\forall \hat{\xi}}} f(y, z) - w + \beta(1 - \delta)\mathbb{E}\left[\left(1 - \lambda_e \tilde{p}\left(\hat{\psi}, \hat{V}(\hat{\xi})\right)\right) J(\hat{\xi}, \hat{V}(\hat{\xi}))\right] \tag{6}$$

subject to the promise keeping constraint

$$V = u(w) + \beta \mathbb{E}\left[\delta U(\hat{\psi}) + (1 - \delta)\left(\hat{V}(\hat{\xi}) + \lambda_e R(\hat{\psi}, \hat{V}(\hat{\xi}))\right)\right]. \tag{7}$$

The match ends only through exogenous separation, with probability δ, or a job-to-job move, with probability $\lambda_e \tilde{p}$; workers cannot quit into unemployment.[15]

Combining the first-order conditions yields, for each future state $\hat{\xi}$,

$$\frac{1}{u'(\hat{w})} - \frac{1}{u'(w)} = -\frac{\partial \log \tilde{p}(\hat{\psi}, \hat{V}(\hat{\xi}))}{\partial \hat{V}} \frac{\lambda_e \tilde{p}(\hat{\psi}, \hat{V}(\hat{\xi}))}{1 - \lambda_e \tilde{p}(\hat{\psi}, \hat{V}(\hat{\xi}))} J(\hat{\xi}, \hat{V}(\hat{\xi})), \tag{8}$$

which, as described in Balke and Lamadon (2022), captures the trade-off the firm faces in changing the wage between two consecutive periods. The left-hand side is the cost: because the worker is risk averse, a stable wage spares the firm the risk premium that wage fluctuations would require. The right-hand side is the benefit: by raising the promise, and hence tomorrow's wage through the envelope condition, in states where the match is valuable, the firm lowers the probability that the worker is poached and retains the associated profits. A firm receiving a favorable productivity shock thus promises more in that state to keep its worker, while one

[15] This omitted margin is empirically small: over 2013–2015, voluntary quits into unemployment accounted for roughly 0.3 percent of employees per year in Sweden (Inspektionen för arbetslöshetsförsäkringen, 2016), against a flow from employment into unemployment of about 5 percent per year in the calibration, so a version of the model matching this flow would have the quit constraint bind rarely.

receiving an adverse shock lets the promise fall. The equation also pins down the sign of wage changes: since $\tilde{p}$ is decreasing in $\hat{V}$ and the middle term on the right-hand side is positive, the wage rises or falls with the sign of J tomorrow. Finally, it also makes clear that on-the-job search is what generates wage fluctuations: without it, the right-hand side vanishes and the firm provides full insurance.

Market tightness. In the search stage, a firm chooses how many vacancies to create and in which submarket. Creating a vacancy in submarket x when the aggregate state is ψ costs k and yields $q(\theta(\psi, x))J(\psi, z_0, x)$: the probability of filling it times the value of a match at the entry productivity z_0 with promise x. With free entry, firms create vacancies in any submarket where this benefit exceeds the cost until the two are equalized, and none where the cost exceeds the benefit. In any submarket visited by a positive measure of workers, tightness is therefore consistent with firms' optimal behavior if and only if

$$k \geq q(\theta(\psi, x))J(\psi, z_0, x), \tag{9}$$

and $\theta(\psi, x) \geq 0$ with complementary slackness. In markets not visited by any worker, consistency requires $q(\theta(\psi, x))J(\psi, z_0, x)$ to be smaller or equal than k. However, following Menzio and Shi (2010), attention is restricted to equilibria in which the following slackness condition is satisfied in every submarket. Note that, since J is decreasing in the promise, condition (9) implies the existence of a threshold $\tilde{x}(\psi)$, defined by $J(\psi, z_0, \tilde{x}(\psi)) = k$, such that $\theta(\psi, x) = 0$ for all $x \geq \tilde{x}(\psi)$: no submarket promising more than $\tilde{x}(\psi)$ is active.

Equilibrium. A recursive equilibrium is a market tightness function $\theta : \Psi \times X \to \mathbb{R}_+$, a return to search function $R : \Psi \times X \to \mathbb{R}$, a search policy function $x^* : \Psi \times X \to X$, a value function for unemployment $U : \Psi \to \mathbb{R}$, a firm value function $J : \Psi \times Z \times X \to \mathbb{R}$, a wage policy function $w : \Psi \times Z \times X \to \mathbb{R}$, a future promised expected utility function $\hat{V} : \Psi \times Z \times X \to X$ and a law of motion for the aggregate state $\Phi_\psi : \Psi \to \Psi$ such that:

- θ satisfies condition (9) for all $(\psi, x) \in \Psi \times X$;
- R solves problem (4), with x^* the associated policy, for all $(\psi, V) \in \Psi \times X$;
- U satisfies equation (5) for all $\psi \in \Psi$;
- J solves problem (6), with w and $\hat{V}$ the associated policies, for all $(\psi, z, V) \in \Psi \times Z \times X$;
- Φ_ψ is consistent with the policies $(x^*, w, \hat{V})$ and the exogenous processes for s and z.

In this definition, $\{\theta, R, x^*, U, J, w, \hat{V}\}$ are in principle functions of the distribution of workers across employment states, (g_u, g_e), which would make the equilibrium demanding to compute. As shown by Menzio and Shi (2010), however, directed search implies that if an equilibrium exists in this environment, it is block recursive: the equilibrium objects depend on the aggregate state ψ only through the state of nature s, and not through (g_u, g_e), which considerably simplifies the solution of the model, described in Appendix E.

4 Calibration and estimation

The model is calibrated at the quarterly frequency.[16] As in Balke and Lamadon (2022), I use a discount rate equal to 0.987 to match an annual interest rate of approximately 5 percent, CRRA utility function $u(x) = (x^{1-\sigma} - 1)/(1-\sigma)$ with $\sigma = 1.5$ and the production function $f(y, z) = e^{y+z}$. The matching function is the one popularized by den Haan et al. (2000), $q(\theta) = (1 + \theta^{\gamma})^{-1/\gamma}$, where I set $\gamma = 0.5$, in between the values used by Hagedorn and Manovskii (2008) and Balke and Lamadon (2022).[17]

For the exogenous processes I proceed as follows. Consistent with the business cycle definition used in the empirical analysis, the aggregate state of nature s is modeled as a two-state (recession R and non-recession NR) first order Markov process. The two probabilities of remaining in the current state between two consecutive periods $\pi_{\mathrm{R}}, \pi_{\mathrm{NR}}$ are calibrated to match the frequency and expected duration of recessionary states in my sample. Specifically, the formulas for the stationary probability of the recessionary state $\pi_{\mathrm{R}}^{\infty}$ and the expected duration of a recession $\mathbb{E}[\mathrm{R}]$ are:

$$\pi_{\mathrm{R}}^{\infty} = \frac{1 - \pi_{\mathrm{NR}}}{2 - \pi_{\mathrm{R}} - \pi_{\mathrm{NR}}}, \qquad \mathbb{E}[\mathrm{R}] = 1 \cdot (1 - \pi_{\mathrm{R}}) + 2 \cdot \pi_{\mathrm{R}} \cdot (1 - \pi_{\mathrm{R}}) + \cdots = \frac{1}{1 - \pi_{\mathrm{R}}}$$

Thus, π_{R} can be calibrated to match the expected length of a recession and, given π_{R}, π_{NR} to match the fraction of time that the economy spends in the recessionary state. In my baseline analysis I consider just the financial crisis as recession. According to the Economic Tendency Index (Appendix C), the period in which growth was much weaker than normal is 2008Q3-2009Q3, therefore, the frequency of recessions in Sweden over the time period 2004-2018, $\pi_{\mathrm{R}}^{\infty}$, is 5/60 and the average length of a recession, $\mathbb{E}[\mathrm{R}]$, is 5 quarters. Using the formulas above, this delivers $\pi_{\mathrm{R}} = 0.8$ and $\pi_{\mathrm{NR}} = 0.982$.

I assume that firm productivity, $\tilde{z}_{jt}$, is the sum of an aggregate and an individual-specific component, respectively y_t and z_{jt}. The aggregate component moves with the aggregate state of nature: log aggregate productivity takes one of two values, y_{R} in recessions and y_{NR} otherwise, with $y_{\mathrm{NR}} > y_{\mathrm{R}}$. I normalize the unconditional mean of productivity in levels to one, $(1 - \pi_{\mathrm{R}}^{\infty})\, e^{y_{\mathrm{NR}}} + \pi_{\mathrm{R}}^{\infty}\, e^{y_{\mathrm{R}}} = 1$, so that the two values are pinned down by the ratio $e^{y_{\mathrm{R}}}/e^{y_{\mathrm{NR}}}$. I calibrate this ratio internally to 0.77 so that the model reproduces the gap between quarterly U2E rates in recessions and normal times (0.199 and 0.245, respectively) in Sweden's Labour Force Survey (LFS), implying $e^{y_{\mathrm{R}}} = 0.785$ and $e^{y_{\mathrm{NR}}} = 1.020$.[18] As is well known since Shimer

[16] All model results in the following sections are obtained on simulated data aggregated to the annual frequency. Appendix E.3 describes the simulation and annualization procedure.

[17] This value generates in the model an average elasticity of the job finding probability with respect to the tightness faced by the unemployed equal to 0.52, in line with the range of values found in the literature (Menzio and Shi, 2011; Petrongolo and Pissarides, 2001; Shimer, 2005).

[18] To capture the slow-moving dynamics of the labour market, I target the 2-quarter lead of the reported U2E rates; the 1-quarter lead yields similar numbers, but the 2-quarter lead better captures the low job-finding rates that follow recessionary periods. Additionally, in order to match the time period of the empirical analysis, I compute

(2005), job-market flows respond little to productivity movements of the size observed in the data. Ljungqvist and Sargent (2017) show that proposed resolutions of this puzzle operate by compressing the fundamental surplus — the share of match output available for vacancy creation. Rather than adopting one of these microfoundations, I take a reduced-form approach: y_R stands in for the composite cyclical forces (profitability, discounting, financial conditions, etc.) that depress the value of a match in downturns, which is why the calibrated productivity decline exceeds typical output drops in recessions.

The firm-specific component, instead, is modeled as an AR(1) process, $z_{jt} = \rho_z z_{jt-1} + \varepsilon_{jt}$ with $\varepsilon_{jt} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_\varepsilon^2)$, with $(\rho_z, \sigma_\varepsilon)$ estimated via indirect inference to match the unconditional variance of $\tilde{z}$ and ν found in the data (0.366 and 0.077, respectively).[19]

As for the remaining parameters, I proceed as follows. As in Balke and Lamadon (2022), firm productivity at the start of the match is set to its median value, $e^{z_0} = 1$. The unemployment benefit b is set to 0.95 so that it equals 70% of the average output of formed matches in the model. In turn, such value also corresponds to about 75% of the average match wage, in line with Swedish replacement rates and in the middle of the range of opportunity-cost estimates obtained by Chodorow-Reich and Karabarbounis (2016) on US data. I set the vacancy cost k to match the unconditional average quarterly U2E rate from the Labour Force Survey (LFS), which in my sample is 24.1%. From the same data, I also recover an unconditional average E2U rate of 1.3% which pins down the separation rate δ. Finally, I normalize the search intensity of unemployed workers to one and set that of the employed ones, λ_e, to 0.37, targeting the quarterly J2J rate of 0.026 reported by Balke and Lamadon (2022) for Sweden.[20]

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labour market flows statistics using the data for the period 2005-2018. See Appendix B for details.

[19] Because the model does not feature firm fixed effects or sector-by-year interactions, this formulation assumes that the volatility captured by the time-by-sector fixed effects in the data is fully accounted for by y_t, and that the variance attributed to firm fixed effects is absorbed by the persistence of z_{jt}. Appendix F describes the estimation procedure more in detail.

[20] J2J series from the LFS are not publicly available for my sample period. Note that this number implies an annual J2J rate of roughly 10%, which is consistent with the fraction of switchers in my sample depicted in Figure A.2.

Parameter	Description	Value	Source / Moment	Target || Model
β	Discount factor	0.987	Balke and Lamadon (2022)	
σ	CRRA coefficient	1.5	Balke and Lamadon (2022)	
γ	Matching function curvature	0.5	Preset	
b	Unemployment benefit	0.95	Benefit-to-avg output ratio	0.7 || 0.699
λ_u	Search intensity unemployed	1	Normalization	
λ_e	Search intensity employed	0.37	J2J rate	0.026 || 0.025
δ	Exogenous separation rate	0.013	E2U rate	0.013 || 0.013
k	Vacancy posting cost	0.16	U2E rate	0.241 || 0.246
π_{R}	Prob. of remaining in R	0.8	Exp. duration recession	5 qtrs.
π_{NR}	Prob. of remaining in NR	0.982	Freq. of recessions	5/60 qtrs.
$e^{y_{\mathrm{R}}}$	Aggregate prod. in R	0.785	$\mathrm{U2E_R}, \mathrm{U2E_{NR}}, \mathbb{E}[e^y]$	0.199 || 0.195, 0.245 || 0.251, 1
$e^{y_{\mathrm{NR}}}$	Aggregate prod. in NR	1.020		
e^{z_0}	Initial firm productivity	1	Balke and Lamadon (2022)	
ρ_z	Persistence of z	0.952 (0.000)	$\mathbb{V}(\tilde{z}), \mathbb{V}(\nu)$	0.366 || 0.366, 0.077 || 0.077
σ_ε	S.d. of innovation to z	0.190 (0.000)		

Table 4: Model parameters. The "Source / Moment" column reports the source for externally set parameters and the targeted moment for calibrated ones; the last column reports the targeted value and the model-generated value (target || model). Employment flow rates are quarterly. Standard errors for the parameters estimated by indirect inference, in parentheses, are computed from 1,000 bootstrap repetitions.

5 Model results

This section presents the quantitative results. I first illustrate the mechanism through which the model transmits idiosyncratic firm shocks to wages and then assess its quantitative fit to the empirical pass-through patterns.

5.1 Inspecting the mechanism

Before presenting the main results, it is useful to build some intuition on the mechanisms through which firm shocks are transmitted to wages in the model. Recall that each match pairs a risk-neutral firm with a risk-averse worker, and that the firm is committed to providing a predetermined amount of promised utility V, by choosing how much to deliver today through the wage w and how much through the state-contingent promised utilities $\hat{V}$ for tomorrow. Absent on-the-job search, the optimal contract in this setting prescribes a constant wage: by fully insuring the worker against idiosyncratic shocks, the firm avoids paying the compensation that risk aversion would otherwise require for bearing wage fluctuations, and thereby minimizes labour costs.

To see how on-the-job search breaks this result, note that since commitment is one-sided, the promised utilities chosen for tomorrow also determine the worker's probability of leaving, so that the intertemporal trade-off now requires the firm to weigh insurance against retention, as described by equation (8). To illustrate the point, consider a firm hit by a negative idiosyncratic productivity shock. Because the shock reduces the value of the match, retaining the worker becomes less valuable, and the firm optimally chooses a low promised utility in that state of the world, which raises the probability that the worker is poached and, through the envelope condition $\partial J/\partial V = -1/u'(w)$, implies a lower wage in that state. The left panel of Figure 6

shows this logic at work in both aggregate states: the average change in promised utility increases with the firm shock, and is markedly negative for adverse ones.[21] The response is asymmetric, however, with promised utility reacting less to positive than to negative shocks. This stems from the no-poaching threshold $\tilde{x}$: as promised utility approaches the level at which no submarket is active, the retention motive drops out and only the insurance one remains, so that favorable shocks move promised utility, and hence wages, by less than adverse ones. The right panel of Figure 6 confirms this reading: for firms hit by favorable shocks — which would raise the promise to retain the worker — average promised utility is already close to or above the threshold,[22] so that the poaching risk is low and the pass-through of positive shocks reduced.[23]

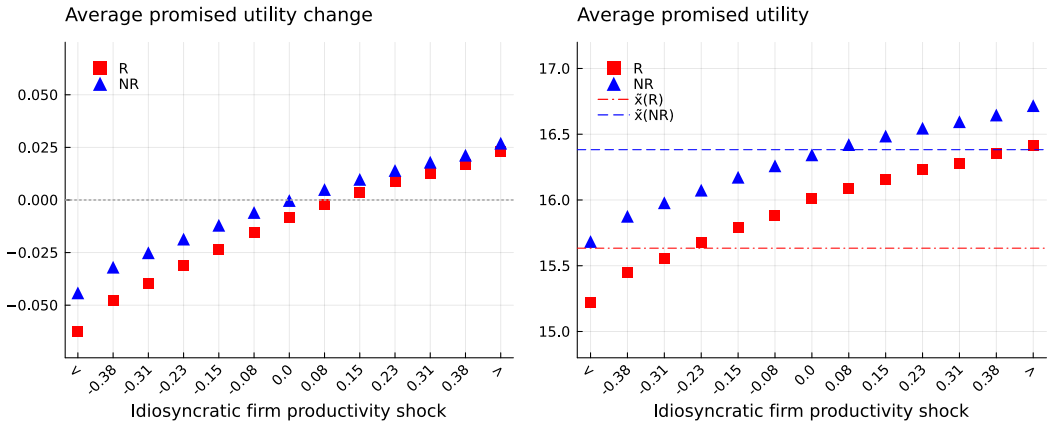


Figure 6: Promised utility by firm productivity shock. Binned means of the relative change in the worker's promised utility, $(V_t - V_{t-1})/V_{t-1}$ (left), and of its level V_t (right), against the idiosyncratic firm productivity shock ν, in the simulated model, for recession (squares) and non-recession (triangles) periods. Dash-dotted (R) and dashed (NR) lines in the right panel mark the no-poaching threshold $\tilde{x}$ in each state. Sample: stayers with an estimated firm shock; shocks trimmed at the 0.5th and 99.5th percentiles within each year. Edge bins collect shocks beyond the outermost interior bins.

Business cycles, in turn, affect the pass-through via a general equilibrium channel that operates differently for positive and negative shocks. For a firm exposed to an adverse shock, lower job-finding rates in recessions mean that inducing the same probability that the worker leaves requires a larger cut in promised utility, and hence in wages, than in normal times — the more so as lower aggregate productivity depresses the value of the match. For a firm hit by a favorable shock, the same logic would predict the opposite, yet the left panel of Figure 6 shows that this is not the case: as seen above, such firms already sit close to or above the no-poaching threshold, and the right panel shows that this holds in both aggregate states. The threshold falls

[21] Firm shocks are computed following the same procedure as in the empirical part, after having annualized the simulated data. More details are reported in Appendix E.3.

[22] Average promised utility can exceed the threshold, since $\tilde{x}$ bounds the set of active submarkets, not the promises a firm can make: once retention is no longer a concern, the firm might still find it optimal to raise promised utility to smooth the worker's consumption.

[23] This reading is also in line with a pattern in the data. Since the cap binds only for workers already paid above what entrants can offer, raises after favorable shocks should be larger for workers whose pay is close to entry offers — arguably those at the bottom of the earnings distribution — and small for everyone else. Consistent with this, the pass-through of favorable shocks is larger in the lowest lagged earnings quintile and flat above it (Appendix Figure D.6).

with aggregate productivity in recessions, but most of the promises incumbents carry into a downturn were made in normal periods, where the economy spends most of its time, so the gap to the bound is wider there. Since the retention motive is the channel through which the aggregate state enters the firm's optimality condition, and it is inactive above the threshold, the cyclicality of the pass-through for positive shocks is effectively removed.

Two facts support this mechanism as a plausible representation of actual firm behavior, especially for negative shocks. First, the mechanism implies that the probability that the worker switches job should decrease with the size of the idiosyncratic firm shock, a prediction directly testable in the data, and consistent with the negative elasticity between the job-to-job rate and firm idiosyncratic shocks that Souchier (2023) reports for France. To this end, Figure 7 plots the model-implied fraction of workers who change employer between $t-1$ and t across bins of the origin firm's shock in year t against its empirical counterpart. Overall, the framework reproduces the data well for negative shocks — note that only the aggregate job-to-job and job-finding rates were targeted in the calibration, not their relationship with firm shocks — while it generates too few switchers for large positive ones. The shortfall there is to some extent expected, as the only source of mobility in the model is idiosyncratic firm shocks, and poaching firms are entrants whose productivity cannot exceed the median z_0. Importantly, however, the missing mobility is a matter of level, not of cyclicality: whatever generates the additional job-to-job moves in the data must operate similarly in recessions and non-recessions, since the switching probability displays no cyclical gap at any point of the shock distribution.[24]

[24] A natural extension would be to allow entrants to have heterogeneous initial productivity, which could sustain poaching of workers at thriving firms and would likely bring the model's switching rates for large positive shocks closer to the data at the cost of a more involved numerical solution. The mechanism behind the acyclicality would remain in place: free entry would still define a no-poaching threshold in each aggregate state, now set by the best available poacher, and the response to favorable shocks would stay acyclical as long as most firms hit by such shocks sit close to or above that threshold in both states. The absence of any cyclical difference in switching, even at the largest favorable shocks, is consistent with this interpretation.

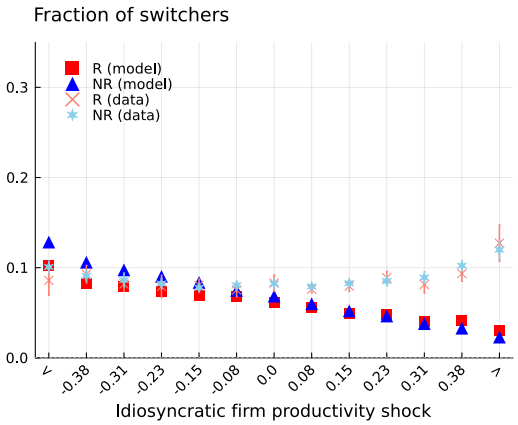


Figure 7: Job switching by firm productivity shock. Fraction of workers who change employer, by bin of the idiosyncratic productivity shock of the firm in the transition year, in the simulated model (filled markers: squares for recessions, triangles for non-recessions) and in the data (light markers: crosses for recessions, stars for non-recessions, values are the same reported in Appendix Figure A.2) The business cycle state is measured in the transition year. For workers who switch, the origin firm's productivity is reconstructed by annualizing the realized idiosyncratic draw of the departing match in the quarter of separation. Workers exiting the sample are excluded when computing the fraction. Vertical bars around the data estimates denote 90% confidence intervals based on standard errors clustered at the firm level. Shocks trimmed at the 0.5th and 99.5th percentiles within each year. Edge bins collect shocks beyond the outermost interior bins.

Second, the mechanism through which firms respond to negative shocks should be particularly relevant in Sweden, which features stricter employment protection than the average OECD country (OECD, 2021). In such a setting, firms are likely to rely on instruments other than dismissal — such as the incentive-based mechanism analyzed in this paper — to adjust their workforce.[25] Consistent with this argument, Figure 5 shows that the business cycle asymmetry for negative shocks is stronger for workers with longer tenure and higher earnings, who are arguably costlier to dismiss.

The next section examines the quantitative implications of the mechanisms just described for the transmission of firm shocks to wages.

5.2 Pass-through over the business cycle

To visualize the effects of the mechanisms described in the previous section on workers' compensation, the left panel of Figure 8 plots the coefficients from a regression of earnings changes on a full set of firm-shock bin dummies, with the zero-shock bin omitted so that each coefficient measures the average earnings change relative to that bin.[26] For positive shocks, the model reproduces both the magnitude of the earnings responses relative to the zero bin and, importantly, the absence of any gap between recessions and non-recessions found in the data. For adverse ones, instead, it generates a larger pass-through than in the data — unsurprisingly, as it abstracts from forces, such as downward wage rigidities, that limit earnings declines — but delivers a gap between the two aggregate states in line with the empirical one.

[25] Strict employment protection also makes the model's one-sided commitment, with the firm bound to the relationship and the worker free to leave, a natural description of the Swedish contracting environment.

[26] The point of the figure is to compare slopes, not levels: the model abstracts from forces that generate positive average earnings growth in the data, and thus cannot match the levels shown in Figure 2.

As the central test of the model lies in its ability to reproduce the shape of the cyclical gap in earnings changes, the right panel of the same figure directly plots the difference between the two states across the distribution of firm shocks in the model and in the data.[27] In both, the gap is close to zero for favorable shocks and widens as shocks become more adverse. Remarkably, this close fit arises from the two forces described in Section 5.1, given that none of the empirical asymmetries was targeted in the calibration.

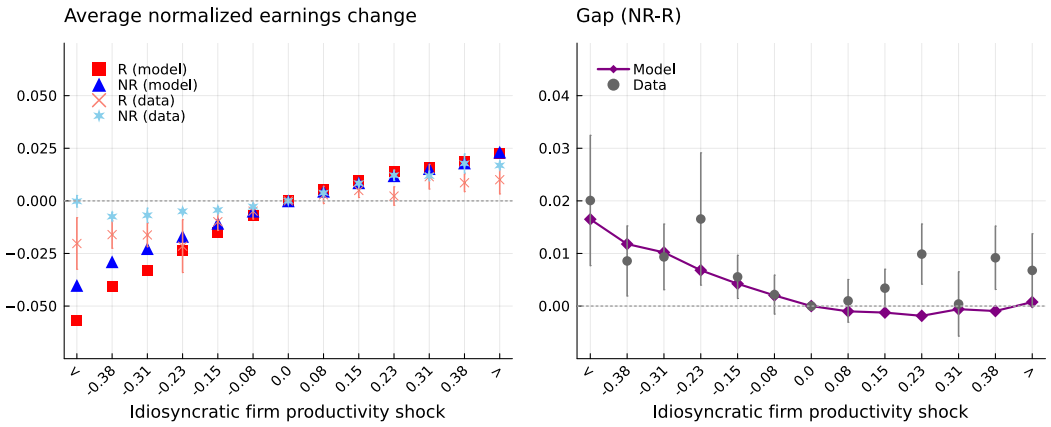


Figure 8: Normalized earnings change and gap by firm productivity shock. Binned means of the average earnings change (normalized to the zero bin, left) and of its gap (right) between non-recessions and recessions, against the idiosyncratic firm productivity shock ν, in the simulated model (diamonds for gap, squares for recessions, triangles for non-recessions) and in the data (circles for the gap, crosses for recessions, stars for non-recessions). Vertical bars around the data estimates denote 90% confidence intervals based on standard errors clustered at the firm level. Sample: stayers with an estimated firm shock; firm shocks and wage changes trimmed at the 0.5th and 99.5th percentiles within each year. Edge bins collect shocks beyond the outermost interior bins.

Table 5 quantifies these patterns by estimating equation (2) on the simulated data, side by side with the empirical estimates from Table 2.[28] The acyclicality of positive shocks' pass-through is confirmed by the regression estimates: the model delivers 0.044 and 0.045 in recessions and non-recessions, in the same ballpark as the empirical 0.032 and 0.039, and as in the data, equality across states is not rejected. The estimates likewise confirm the patterns for the negative side: levels are higher than in the data (0.117 and 0.082 vs. 0.029 and 0.015), reflecting, as previously mentioned, the absence of forces limiting earnings declines, but the model reproduces the cyclical asymmetry, with the equality test rejecting the null of equal coefficients across states.

[27] The gap is estimated from a pooled regression of earnings changes on shock-bin dummies fully interacted with a non-recession indicator, in which the zero-bin interaction is replaced by a common non-recession shifter, so that each plotted coefficient measures the cyclical gap in a given bin relative to the gap at the zero-shock bin.

[28] The regression on simulated data mirrors the empirical specification, except that controls with no model counterpart are omitted. The estimates are essentially unchanged when the included controls are also dropped.

Dependent variable: log earnings change				
	Recession		Non-recession	
	Data	Model	Data	Model
ν_{jt} (β_1)	0.032***	0.044***	0.039***	0.045***
	(0.005)	(0.001)	(0.002)	(0.000)
$\nu_{jt} \cdot \mathbb{1}\{\nu_{jt} < 0\}$ (β_2)	−0.003	0.073***	−0.024***	0.037***
	(0.011)	(0.002)	(0.003)	(0.000)
Sum: $\beta_1 + \beta_2$	0.029***	0.117***	0.015***	0.082***
	(0.008)	(0.001)	(0.002)	(0.000)
Equality Tests (p-value)	Data		Model	
$H_0 : \beta_1^R = \beta_1^{NR}$	0.149		0.268	
$H_0 : (\beta_1 + \beta_2)^R = (\beta_1 + \beta_2)^{NR}$	0.091		0.000	

Table 5: Pass-through of firm shocks: model and data. Estimates of the asymmetric specification of equation (2) in recession and non-recession years, in the data (from Table 2) and in the simulated model, where the regression includes the controls of equation (2) that have a counterpart in the model. Standard errors clustered at the firm level in parentheses; $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. The equality tests report p-values for the difference between recession and non-recession coefficients, obtained from a pooled specification interacting the regressors with the business cycle state. Both samples trim earnings changes and shocks at the 0.5th and 99.5th percentiles within each period.

6 Implications

Having established that the model reproduces the cyclical pattern of pass-through documented in the data, in this section I show that the mechanisms previously described have implications beyond this fit. Specifically, the model generates the cyclical behavior of income risk documented in the data, and allows me to study how public policy — in the form of unemployment benefits and hiring subsidies — can shape the insurance that firms privately provide to their workers.

6.1 Cyclical income risk

In a seminal paper, Guvenen et al. (2014) show that the distribution of earnings changes in the US features procyclical skewness and acyclical variance. Busch et al. (2022) document the same patterns in Sweden (as well as in Germany and France), and show that the skewness result survives when conditioning of full-time workers, indicating that the extensive margin is not its only determinant.

In principle, the insurance mechanism underlying the transmission of firm shocks described in the previous sections could generate these regularities: if firms pass through negative shocks more strongly in recessions while the transmission of positive ones is acyclical, the left tail of wage changes thickens in downturns, lowering skewness while leaving the standard deviation largely unchanged. Figure 9 investigates whether this is the case by showing the simulated distribution of log wage changes among stayers[29] before, during, and after a recessionary

[29]The figure includes all stayers, regardless of whether the firm-shock measure ν used in the regression analysis is available for the match, consistent with the samples of the empirical income-risk literature. Restricting to

episode.[30]

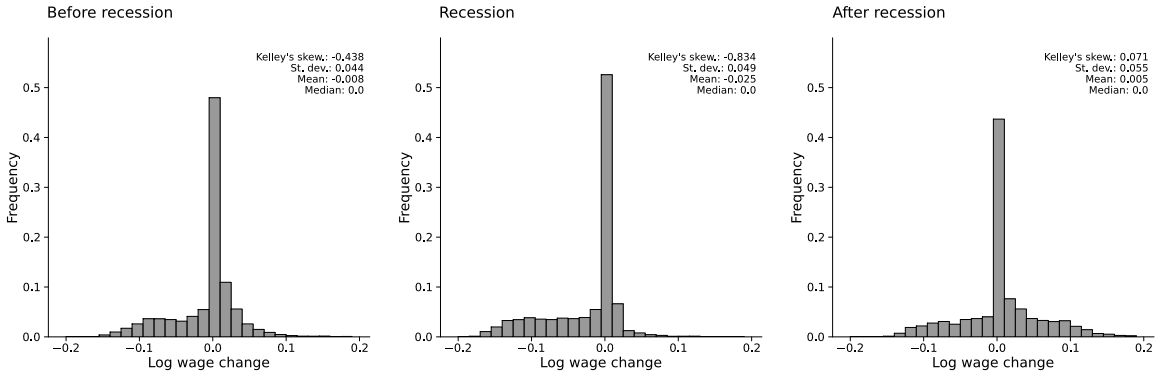


Figure 9: Distribution of wage changes over a recessionary episode. Histograms and summary statistics of log wage changes among stayers in the simulated model, in the year before the recession (left), during the recession (center), and in the year after (right).

Overall, the model successfully reproduces the two empirical patterns. During the recession, firms cut wages more, which shifts mass into the negative region of the distribution, thereby lowering skewness from -0.438 to -0.834. During the recovery, firms' renewed incentive to retain workers at the higher aggregate productivity level generates many positive wage changes, which push skewness up to 0.071. The standard deviation, by contrast, barely moves. In the downturn, wage cuts become more frequent but not deeper, and positive changes — already small as they are bounded by the no-poaching threshold — become rarer, so all the reshuffling occurs in regions to which dispersion is largely insensitive. The slight increase when the economy returns to normal times, instead, reflects firms reversing the wage cuts made during the recession. These raises differ in size across matches, depending on how much the wage was pushed down, and this heterogeneity briefly widens the distribution.

Three further features of the figure are worth noting. First, the majority of stayers face essentially no wage change at all, indicating that the insurance mechanism in the calibrated model is strong. Second, as shown by the before-recession panel and by the time series in Appendix Figure A.4, Kelley's skewness in normal times stabilizes at a negative value, which follows from the fact that negative changes are not bounded while positive ones are limited by the no-poaching threshold. Finally, while the aim of the exercise is qualitative, the skewness levels are within the range of values reported in Figure 2 of Busch et al. (2022) for Sweden.[31]

Taking stock, this section has shown that the insurance forces governing the firm-worker relationship can account for part of the cyclicality of income risk. Policies aimed at addressing

observations with non-missing ν leaves the patterns essentially unchanged.

[30] For the sake of the exposition, the figure plots three consecutive years: the first in normal times, the second in recession, and the third back in normal times. This window is representative: since the aggregate state follows a two-point Markov chain, adjustments are concentrated around transitions, and the same patterns repeat at every recessionary episode in the full simulated series, reported in Appendix Figure A.4, with magnitudes slightly varying with the timing and length of each recession.

[31] The standard deviation is instead substantially lower than their estimates, consistent with the fact that the model abstracts from other sources of earnings volatility, such as worker-level shocks.

this risk can therefore also operate through this channel. The next section investigates this possibility, studying whether unemployment benefits and the cost of job creation affect the protection firms provide, and how this depends on the state of the economy.

6.2 Can public policy affect firm-provided insurance?

This section examines how the insurance firms provide to their workers responds to two margins through which governments can intervene in this economy: the generosity of unemployment benefits and the cost of job creation.[32] A priori, the impact of either instrument on the employment contract is ambiguous. A higher benefit b may crowd out firm-provided insurance, but it also lowers the match surplus and thus firms' stake in retaining workers. In turn, a lower cost of job creation k raises the poaching threat through more entry, but also lowers the value of a filled job and thus, as for the unemployment benefit, firms' stake in retaining workers.

To quantify these effects, I solve the model for equally spaced values of the unemployment benefit ranging from $b = 0.35$ to $b = 1.05$ — roughly 35 to 80 percent of average match output, against 70 percent in the baseline — and for values of the vacancy cost ranging from $k = 0.08$ to $k = 0.64$, half to four times its calibrated value, in each case holding all other parameters at their calibrated values. Figure 10 plots the resulting pass-through elasticities for positive (top row) and negative (middle row) idiosyncratic firm shocks against each parameter, together with the job-finding and job-to-job rates (bottom row).

[32] Hiring credits, employer social contributions on new hires, and administrative hiring frictions are examples of levers through which policy can affect the cost of creating a job.

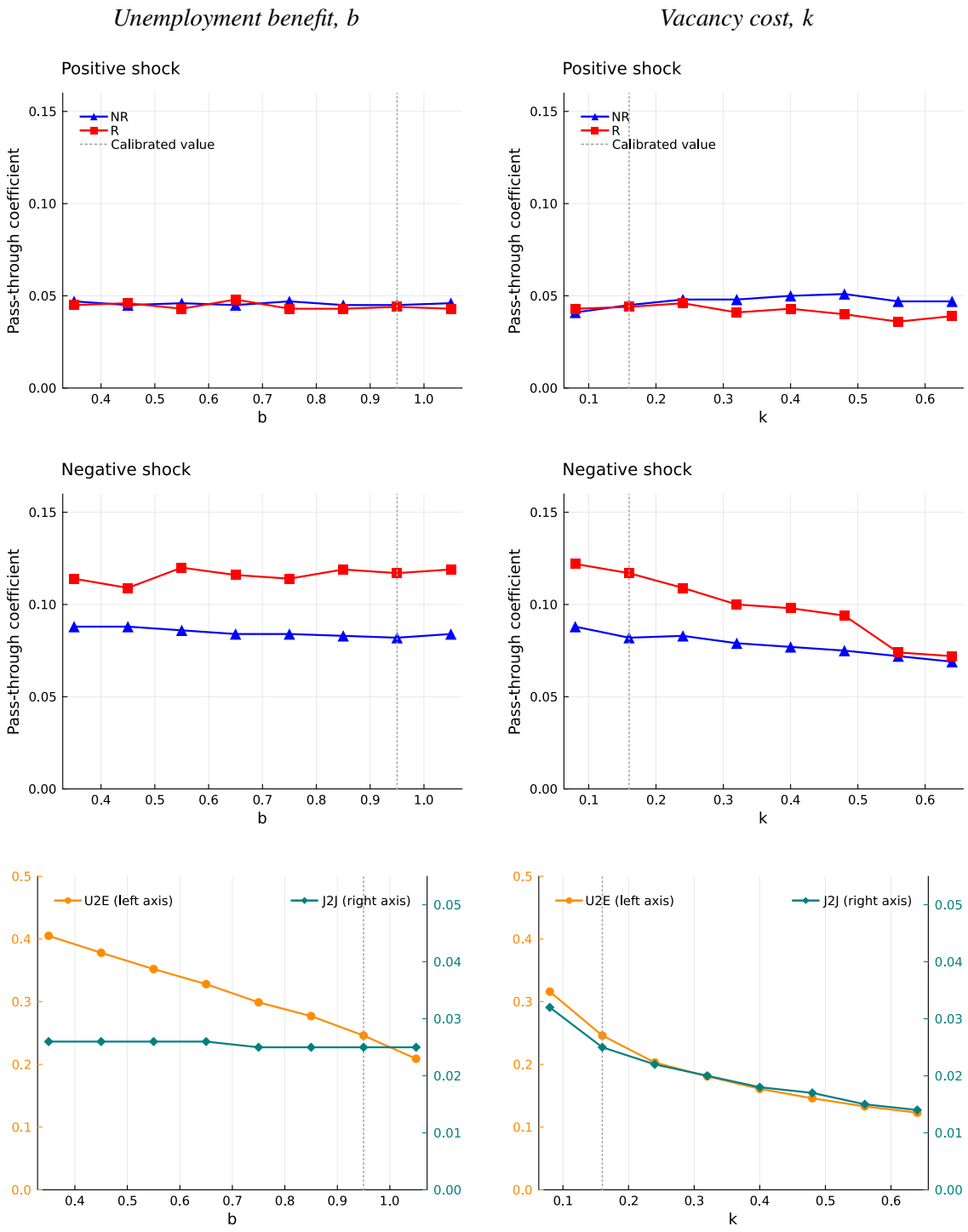


Figure 10: Pass-through and labour market outcomes under alternative parameter values. The left column varies the unemployment benefit b and the right column the vacancy cost k, holding all other parameters at their calibrated values; dashed vertical lines mark the baseline values of $b = 0.95$ and $k = 0.16$. The top and middle rows report pass-through elasticities of positive and negative firm shocks, respectively, in recessions (squares) and non-recessions (triangles), estimated from model-simulated data. The bottom row reports the average job-finding rate (left axis, circles) and job-to-job rate (right axis, diamonds).

The left column shows that both the level of pass-through and its cyclical asymmetry are essentially invariant to b over the entire range considered. Public unemployment benefits thus neither crowds out nor crowds in the insurance firms provide in this model. The right column, by contrast, reveals that pass-through does respond to the cost of job creation. The elasticity

of negative shocks declines as job creation becomes more costly in both states, but faster in recessions, so that the cyclical gap narrows across the grid and even disappears at high values of k. Positive shocks, on the other hand, are largely unaffected in both states, with only a small and constant gap opening after the lowest values of k.

Why does the benefit leave pass-through untouched while the vacancy cost affects it? The bottom row of Figure 10 points to the answer: both a lower benefit and a cheaper vacancy raise the job-finding rate, but only the latter increases the job-to-job rate, which measures the poaching threat existing matches face and is what matters for pass-through.

The difference reflects where each instrument operates. A lower benefit makes the unemployed search in submarkets that promise less: their outside option has fallen, so lower promises are acceptable and, since those submarkets are tighter, they find jobs faster, which is why the job-finding rate rises as b falls. New matches thus start from lower promises, and the distribution of utilities among the employed shifts left accordingly. Crucially, however, the lower benefit affects the value of a match in essentially the same way along the whole range of promises: it shifts the level of J but not its slope, and it is the latter that pins down wages through the envelope condition. With the wage associated with any given marginal value unchanged, the left-hand side of equation (8) is unaffected, and the condition then requires the poaching probabilities at the chosen contracts to be unaffected as well. Promised utilities thus absorb the change in the outside option and the job-to-job rate does not move, as the flat profile across the b grid confirms. This offers a rationale for the evidence of Jäger et al. (2020) that incumbents' wages are essentially insensitive to unemployment benefits: the latter change what workers are promised, not what they are paid.

The vacancy cost, by contrast, acts on the poaching threat directly. A higher k lowers job-finding and hence the poaching threat in both states, and compresses the difference between them — with expensive job creation, hiring is less viable also in non-recessions, so a recession has little left to take away — which is why the cyclical gap closes. This poaching channel thus complements the one identified by Cahuc et al. (2019), who find in a directed search model without on-the-job search that permanent hiring credits are partly absorbed into wages through the reservation wage of the unemployed.

Two conclusions follow for policy. First, in this model unemployment insurance cannot affect the pass-through of firm shocks: its generosity determines how painful and how long unemployment spells are, but not how much of their employer's shocks workers bear. Second, hiring subsidies make firms worse insurers, and hiring frictions better ones, at the price of less job-to-job mobility and a lower job-finding rate.

7 Conclusion

Using Swedish administrative data covering the universe of non-financial private firms, employment relationships, and workers, this paper has shown that the elasticity of earnings with respect to positive idiosyncratic firm productivity shocks is positive and acyclical, while that of negative shocks is also positive and about twice as large in recessions as in normal times. These patterns survive correcting for selection into staying, alternative definitions of business cycle episodes, and alternative measures of idiosyncratic firm productivity.

A directed search model of the labour market with on-the-job search, risk-averse workers, and firm commitment rationalizes both features. Free entry bounds the utility entering firms can offer, and firms hit by favorable shocks sit close to that bound in both aggregate states, which caps their wage responses over the cycle; depressed hiring in downturns means that inducing the same probability that the worker leaves requires a deeper cut in promised utility, and hence in wages, after adverse shocks. The same mechanisms generate a wage growth distribution among stayers with procyclical skewness, suggesting that the firm insurance channel is a relevant force behind the cyclicality of income risk empirically documented in the literature.

The model also has implications for how public policy interacts with the insurance that firms provide to their workers. In settings with high employment protection and rare voluntary quits, unemployment benefits leave pass-through unaffected: their generosity shifts the level of match values and the promises over which parties contract, but not the wage responses to firm shocks. By contrast, costlier vacancy creation weakens the poaching threat in both aggregate states, thus closing the cyclical gap for negative shocks. The flip side is a policy trade-off: measures that foster labour market dynamism by making job creation cheaper also erode the insurance of incumbent workers, and do so disproportionately in recessions, precisely when that protection is needed the most. Understanding under which circumstances one side of this trade-off should prevail over the other is left to future research.

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Online Appendix for "Business cycle asymmetry of earnings pass-through"

A Additional tables and figures

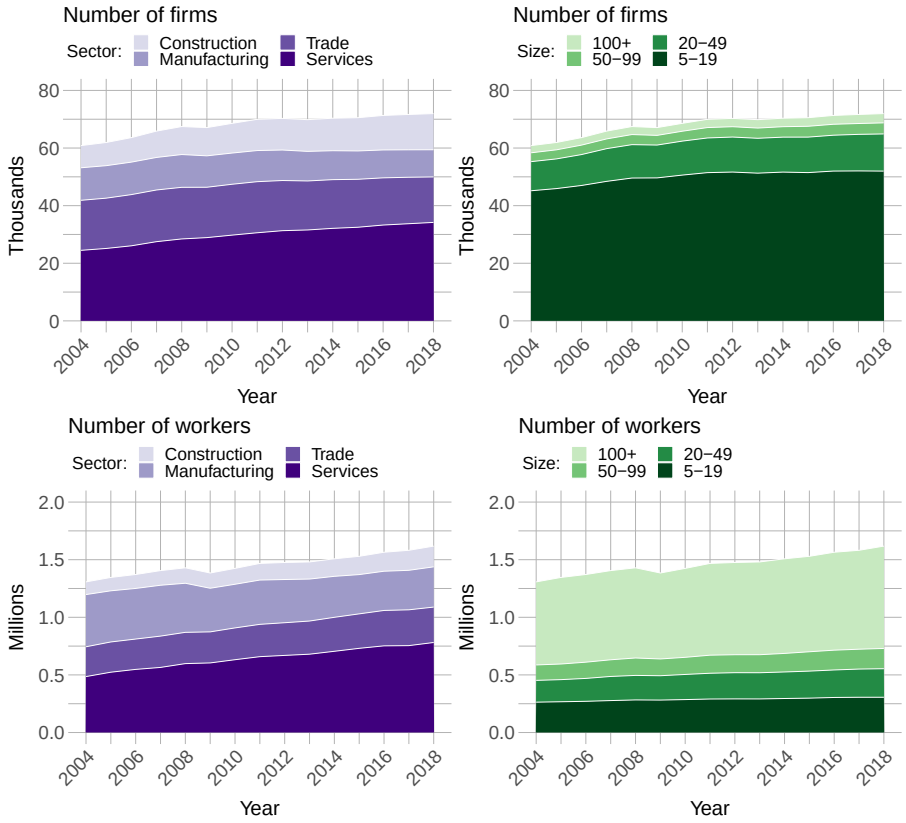


Figure A.1: Sample coverage. Number of firms and workers in the sample by broad sector and firm-size group over 2004–2018.

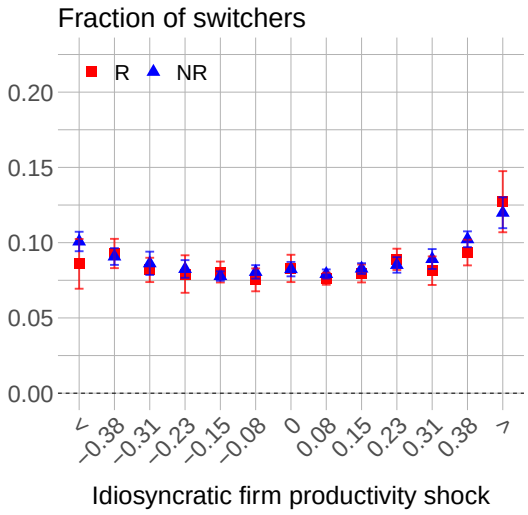


Figure A.2: Job switching and firm shocks. Fraction of workers who change employer by bin of the origin firm's idiosyncratic productivity shock in the transition year, in recessions (squares) and non-recessions (triangles). Workers exiting the sample are excluded when computing the fraction. Vertical bars denote 90% confidence intervals based on standard errors clustered at the firm level. Shocks trimmed at the 0.5th and 99.5th percentiles within each year. Edge bins collect shocks beyond the outermost interior bins.

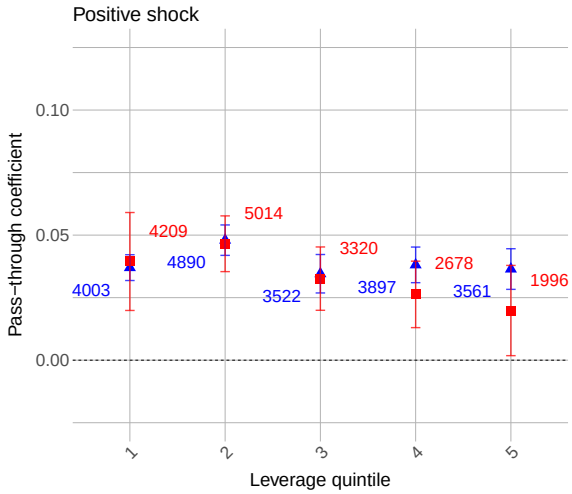


Figure A.3: Pass-through by firm leverage, positive shocks. Pass-through coefficients for positive firm shocks, estimated separately by quintile of lagged firm leverage (quintile 1 = lowest leverage), in recessions and non-recessions. Vertical bars denote 90% confidence intervals based on standard errors clustered at the firm level. Monetary values are expressed in 2022 SEK and computed as the coefficient times the standard deviation of the firm shock times average earnings in each group.

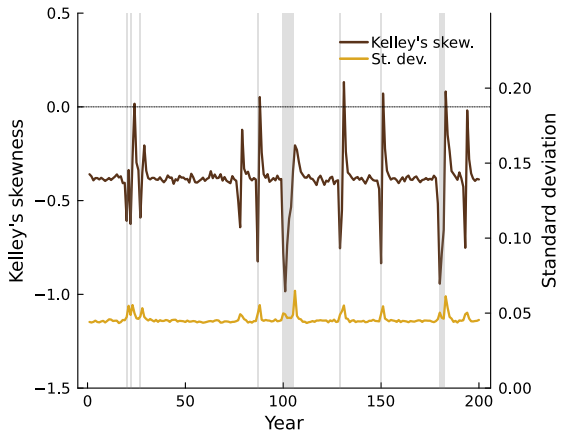


Figure A.4: Cyclicality of income risk in the model. Kelley's skewness (left axis) and standard deviation (right axis) of the cross-sectional distribution of log wage changes among stayers, computed each year from model simulations. Shaded areas denote years classified as recessions.

Dependent variable: $\log \Delta w_{ijt}$, log earnings change, Manufacturing sector		
	R	NR
β_1 ($\nu_{jt} \geq 0$)	0.029^{***}	0.030^{***}
	(0.008)	(0.004)
$\beta_1 + \beta_2$ ($\nu_{jt} < 0$)	0.034^{**}	0.012^{***}
	(0.014)	(0.004)
R^2	0.011	0.008
Obs. (Millions)	0.7	3.6
Equality Tests (p-values)		
$H_0 : \beta_1^R = \beta_1^{NR}$		0.966
$H_0 : (\beta_1 + \beta_2)^R = (\beta_1 + \beta_2)^{NR}$		0.109

Table A.1: Baseline specification restricted to the manufacturing sector. The dependent variable is log earnings change. Firm productivity is measured as value added per worker and idiosyncratic firm shocks are computed using the specification in equation (1). Results are split by recessionary (R) and non-recessionary (NR) years. Standard errors are clustered at the firm level; $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Equality tests are conducted by interacting the specification with business-cycle indicators and testing whether the differential effect across states is equal to zero.

B Data

Structural Business Statistics (FEK). This dataset contains income and financial statement variables, along with industry codes based on the Swedish Standard Industrial Classification (*Standard för svensk näringsgrensindelning*, SNI) for non-financial private Swedish firms.[33] This database has a long history — financial account statistics have been produced since the 1950s — and data collection procedures have undergone several redesigns over the years. The current version of FEK combines the usage of both administrative data and surveys and it has been in use since 2003. A significant enhancement occurred in 1997, when the comprehensive usage of administrative data expanded the coverage to include the universe of firms. Overall, FEK includes all two-digit SNI 2007 codes from 01 to 96, except 64-66 (financial and insurance activities) and 84 (public administration and defense, compulsory social insurance).

I drop observations with missing or non-conforming firm identifier and exclude firms with more than one entry per year. Additionally, I convert SNI codes across all years to be consistent with the SNI 2007 definition.[34] Furthermore, I exclude firms that, even after the conversion, remain classified as financial companies according to their SNI codes. I also construct a leverage measure as the sum of short- and long-term liabilities over total assets, and a measure of tangible capital as the sum of equipment, machinery, building and land reported in the firm's balance sheet. However, since it is known that firms might have incentives not to report the actual value of capital used in production, I also construct tangible capital via the perpetual inventory method using the formula $K_{j,t} = (1-\delta)K_{j,t-1} + I_{j,t}$, where K is real capital, I real net investment (in equipment, machinery, building and land) and δ the depreciation rate, j indicates the firm and t time. Depreciation rates δ at the 2-digit SNI 1992 level are taken from Melander et al. (2009).[35] For each sector, the depreciation rate is a weighted average of the equipment and machinery rate and the building and land rate (the authors set the latter equal to 0.0314) with weights given by the sector-specific shares of these two components in tangible capital, averaged over time. Sectors not covered by Melander et al. (2009) and those with all zero values for equipment and machinery or building and land are excluded. The capital series is initialized in 1997 using the non-negative book value of real tangible capital (and set to missing otherwise). For subsequent years, I apply the formula above if the firm was present in the previous year — if the resulting value is negative I set it to missing and, as Carlsson et al. (2016), if it is below the book value I set it equal to the latter — and for entering firms I use the same procedure adopted for 1997. If a firm has year gaps, I proceed in the same fashion

[33] In this paper, the unit of analysis is the firm, defined as a legal entity that can include one or more establishments.

[34] SNI definitions changed twice in the time range considered. To get a consistent definition of SNI codes across years, I extend the SNI 2007 definition back in time. To do this, I use conversion files provided by Statistics Sweden and, when the conversion is not one-to-one, I use the most common transition in the data in the years around the switch in SNI definitions (if there is a tie, I select the code with the highest total value-added).

[35] This requires constructing conversion tables from SNI 2007 to SNI 2002 and from SNI 2002 to SNI 1992, which I do following the procedure described for the consistent SNI 2007 index in reverse order.

as I did for the initial year for the year they get back in the sample. As a last step, in order to perform the check based on the methodology by De Loecker (2011), I construct a measure of log value-added for each 2-digit sector (according to SNI 2007) by computing, for every year, the weighted sum of firm-specific log value-added with weight being the firm's share of total sales in the sector.[36] Finally, monetary variables are deflated using the GDP deflator for Sweden with reference year 2022.

Register-Based Labour Market Statistics (RAMS). On the firms' side, RAMS includes information on employment, juridic status, sector of activity and municipality at the establishment and firm level from 1985 to 2019. I exclude observations with missing or non-conforming firm identifier. In addition, since municipality is observed at the establishment level,[37] if more than one municipality is associated to the same firm in a given year, I assign to that firm/year observation the municipality where the highest number of employees is located (I randomize in case of ties).[38] From the first two digits of the municipality code computed as just described, I recover the region in which the firm is based.

On the workers' side, RAMS includes for every gainfully employed individual the establishment and firm identifiers of all their workplaces, the starting and ending month of each employment relationship, a measure of the total compensation received in each employment relationship and the occupational status of the worker (sailor, employee, farmer, self-employed, entrepreneur in own limited liability company) for the time period 1985-2019. I exclude observations with missing or non-conforming firm identifier and observations for which the starting month of the employment relationship is zero and the ending month is not. Furthermore, I consider only individuals whose occupation is of employee or entrepreneur with their own limited liability company[39] and who are employed at a company for at least six months in a year.[40] If a worker has more than one employment relationship in a given year, I classify that worker as employed at the firm where they received the highest salary (I randomize in case of ties). Based on this procedure, I construct a tenure indicator that keeps track of the number

[36] To perform this, therefore, I focus on firms with non-negative sales and positive value-added, with both measures deflated with sectoral prices at the 2-digit level.

[37] From 1997 onward, I ensure consistency of the municipality index by harmonizing three cases: Nykvarn (0140) recoded to Södertälje (0181) from 1998; Knivsta (0330) recoded to Uppsala (0380) from 2002; Heby recoded uniformly as 0331 in all years to maintain consistency with region identifiers (switched from 1917 to 0331 in 2006 – region codes are the first two digits) and because most sample years use this code.

[38] There are also very few cases in which a firm is available in the firms' RAMS dataset but not in the workers' one. In that case the procedure cannot be applied and I use the original value reported in the former database.

[39] In RAMS it is not possible to separate the two groups because entrepreneurs with their own limited liability company had the same occupation code as employees until 2010. This is because, similarly to employees, they also get a salary from the company and so they were not considered separately until recent years.

[40] Some individuals in this occupational category have zero as initial and final month of the employment relationship. Upon questioning Statistics Sweden, I got to know that they should be treated as employed for the full year.

of years an employment relationship has been going on,[41] a variable that counts the number of employees (using the definition of employment just decribed) in each firm and an annual measure of earnings by annualizing the total salary received. Monetary variables are deflated using the CPI for Sweden with reference year 2022.

Longitudinal Integrated Database for Health Insurance and Labour Market Studies (LISA). LISA contains socio-economic and demographic information for each Swedish resident for the period 1990-2021. From this dataset, I recover civil status, gender, year of birth and education. The education variable can have the following six levels: (1) pre-secondary education shorter than nine years, (2) pre-secondary education nine years or equivalent, (3) secondary education, (4) post-secondary education shorter than two years, (5) post-secondary education two years or longer, (6) postgraduate. I merge civil statuses to get four categories: singles, married (or in a registered partnership), separated (also from a registered partnership), survivors (also from a registered partnership). For a few individuals, gender and year of birth change over time and the education level is decreasing (these are typically instances in which the data are revised based on updated information). When this happens for gender and year of birth, I use the value in the last year the individual is present in the data. For education, I identify the most recent negative change in education level (if present) and then I substitute the subsequent education level back in time in all cases in which the education level was higher (and non-missing) until I get a non-decreasing series. In addition, if education is the same before and after a block of missing values, I replace the missing values with the education level at the block extrema (and leave missing otherwise, as it is not possible to know exactly when the change happened). Additionally, under the assumption that more recent data is of better quality, if some individual has missing education in the last year she is present, then education is set to missing in all years. Furthermore, in the data there are some civil status transitions which should not be possible: a person cannot return single after having been married or in a registered partnership and it is not possible to enter into a registered partnership after 2009 as the law was abolished. In the former case, I assign the single civil status to all the entries before the switch. Also, if single is the first civil status after missing values, I assign the single civil status to all observations before. In the latter case, I exclude the individuals for whom that happens.[42] Furthermore, after 1998 – where the household identifier variable is more reliable – I recover for married people their spouse individual identifier (I focus on cases in which the household identifier returns only two married people, as the other cases are possibly due to miscoding or individuals not having updated their civil status yet at the moment of data collection). Then, I recover change of spouse event indicators by checking whether the civil

[41] In doing that, I assume that if a worker returns to a company where they have already worked before they resume tenure from where they left.

[42] The civil statuses classification was revised in 1998. While the civil status variable in the data has been translated, values before 1998 seem less reliable.

status indicator and the spouse indicators between two consecutive years have changed (note that, for civil status, this procedure does not record within-year changes and, for both, it does not record changes after gap years – the latter because it is not possible to know exactly when the change happened).

Aggregate data. Time series data for GDP, CPI and the unemployment rate are obtained from the OECD Economic Outlook 113, June 2023 version (the reference year for this dataset is 2022). 2-digits sector prices (PPI) are from Statistics Sweden's website.[43] Total and sectoral value-added and employment at the quarterly frequency are taken from Statistics Sweden's website and then aggregated at the annual frequency by summing up the quarterly flow values in each year for value-added and by averaging across the four quarters for employment. The quarterly real GDP series used to measure the recession productivity gap in Appendix F is from FRED. The Economic Tendency Indicator was downloaded from the website of the Swedish National Institute of Economic Research (*Konjunkturinstitutet*). Quarterly worker job flows come from the Labour Force Survey (package 1 version), and are publicly available at Statistics Sweden's website. The data cover 2005Q3 onwards, and I focus on series referring to individuals aged 25–54. The employment-to-unemployment (E2U) rate is the transition rate from employment (*Sysselsatta*) to unemployment (*Arbetslös*). The unemployment-to-employment (U2E) rate is the sum of transitions from unemployment into permanent employment (*Fast anställd*) and into temporary employment (*Tidsbegränsat anställd*). Because transitions from unemployment into self-employment and assisting family membership are not published with the age breakdown, the employment concepts underlying the two rates differ slightly. The U2E rate covers transitions into permanent and temporary employment, whereas the at-risk group for the E2U rate, (*Sysselsatta*), comprises permanent and temporary employees as well as the self-employed and assisting family members.

Sample restrictions. On the firm side, in addition to the main criteria stated in the text — limited liability companies operating in manufacturing, construction, trade, and services, with at least five employees and positive value added — I retain firms with non-missing municipality code, a positive wage bill, non-negative operating expenditures, non-negative turnover, non-negative tangible capital, positive total assets and total fixed assets, and a liabilities-to-assets ratio between minus one and four.[44] Among the two-digit SNI 2007

[43] I restrict to manufacturing (10-33) as for the other sectors these data are less granular and cover fewer years. Additionally, for some sectors data are incomplete, so I proceed as follows. For sectors 12 and, I use the prices aggregate prices for manufacturing; for sector 14, I use the price index for 14+15.2 as the latter is very close to the price index for 14 in the overlapping years; for sector 21, I use the price index for 21.1 as the two series seems close in overlapping years and I use linear interpolation to get missing price values for 2012 and between 2015 and 2018; for sector 33, I obtain the value for 2004 with linear extrapolation using the 2005-2006 values.

[44] The upper bound for leverage roughly corresponds to the 99.97th percentile of the liabilities-to-assets distribution; the lower bound excludes companies with total credits larger than total assets.

codes covered by FEK, manufacturing corresponds to codes 10–33, construction to 41–43, trade to 45–47, and services to 49–96. I exclude sector 95 before 2006, as total sectoral value added does not match well the aggregate figures reported by Statistics Sweden. On the worker side, in addition to the age restriction (25 to 55 years old) and to considering only individuals whose occupational status is employee, I require non-missing education and civil status. Entrepreneurs with their own limited liability company are included among employees if they appear in the firm's payroll: since RAMS does not distinguish this group from employees before 2010, I keep both groups throughout the sample period to ensure time consistency. To limit the influence of extreme values, I retain observations with annual nominal earnings above a survival threshold and below the 99.999th percentile of the nominal earnings distribution in each year. The survival threshold annualizes the monthly subsistence income levels for single individuals without children published by the Swedish National Board of Health and Welfare (*Socialstyrelsen*, available at www.socialstyrelsen.se/kunskapsstod-och-regler/omraden/ekonomiskt-bistand/riksnormen).

C Robustness

This appendix provides implementation details and full estimates for the robustness checks summarized in Section 2.3.1 of the main text. Tables C.2 and C.3 collect the results.

Business cycle definitions. The baseline analysis classifies 2008–2009 as recessionary years. To assess how the results depend on this choice, I consider two alternative definitions based on the Economic Tendency Indicator (ETI), compiled by the Swedish National Institute of Economic Research (*Konjunkturinstitutet*) from monthly surveys of households and firms. Values above 100 and 110 indicate, respectively, stronger and much stronger economic growth than normal, and values below 100 and 90 weaker and much weaker growth. Figure C.5 plots the indicator at the quarterly frequency, aggregating the monthly data by averaging. The baseline years are the only ones in which the indicator remained much weaker than normal for at least two consecutive quarters, confirming that they capture a particularly deep downturn. The two alternative definitions progressively broaden the classification: the first counts a year as recessionary if the ETI was much weaker than normal in at least one quarter, which adds 2012; the second if it was weaker or much weaker than normal in at least two consecutive quarters, which extends the recessionary period to 2011–2013.

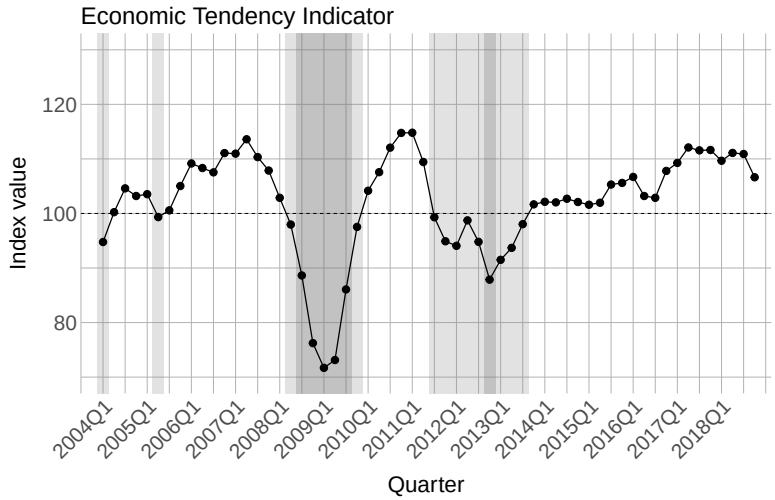


Figure C.5: Economic Tendency Indicator for Sweden. Values above 100 (110) indicate stronger (much stronger) than normal economic activity; values below 100 (90) indicate weaker (much weaker) than normal activity. Light shading marks periods with the indicator below 100; dark shading marks periods below 90.

Columns 1–4 of Table C.2 report the results. The coefficients remain close to the baseline under both classifications, and the equality test for the pass-through of negative shocks continues to reject. The difference in negative-shock pass-through between recessions and non-recessions narrows somewhat as milder years are included, however, indicating that the asymmetry is most pronounced in deep downturns, when unemployment rises and job-finding falls the most — consistent with the model of Section 3, in which the recession excess in pass-through stems from the weaker outside options implied by the gap in job-finding rates between the two states.

Extraction of firm shocks. The baseline specification of equation (1) includes firm fixed effects, which avoids attributing permanent productivity differences to the shock at the cost of possibly introducing Nickell bias (Nickell, 1981). To assess the sensitivity of the results to these issues, I consider three alternative ways of extracting the shocks: a balanced panel of firms, which estimates the fixed effects on the longest available histories; the log difference of productivity as the shock, which removes permanent heterogeneity but includes no other controls; and equation (1) without firm fixed effects, which attributes persistent firm-level differences to the shock.[45] Columns 5–10 of Table C.2 report the estimates. The balanced panel yields coefficients close to the baseline, indicating that changes in the set of firms present across years do not drive the results. The equality test on the negative side does not reject at conventional levels, but this likely reflects the smaller sample, which reduces statistical power. The log-difference shock also preserves the stronger pass-through of negative shocks in recessions, with the equality test p-value just above 10 percent, though the gap between states narrows. On the positive side, the gap is only slightly larger than in the baseline (0.009 versus 0.007), yet the equality test now rejects at the 10 percent level. This possibly reflects the

[45] In the last two cases I do not require at least five usable observations per firm, since the restriction serves to estimate the firm fixed effect, which these specifications do not include.

larger sample, which includes firms observed for as little as two years, for which a long-term insurance relationship is unlikely to have formed. The specification without firm fixed effects departs most from the baseline: the pass-through of negative shocks in non-recessions nearly doubles, from 0.015 to 0.027, closing most of the gap between states. This is consistent with permanent heterogeneity entering the shock, so that persistently low-productivity firms appear to receive negative shocks. Overall, the cyclical asymmetry survives alternative extractions of the shocks, and the one specification in which it weakens materially is the one that does not separate innovations from permanent firm heterogeneity, which supports the baseline choice of including firm fixed effects.

Dependent variable: $\Delta \log w_{ijt}$, log earnings change										
	Alternative Business Cycle Definitions				Alternative Firm Shock Measures					
	ETI very weak ($\geq$ 1 qtr.)		ETI weak/very weak ($\geq$2 cons. qtrs.)		Balanced		Log difference		No firm FE	
	R	NR	R	NR	R	NR	R	NR	R	NR
β_1 ($\nu_{jt} \geq 0$)	0.033***	0.039***	0.036***	0.039***	0.034***	0.044***	0.031***	0.040***	0.037***	0.035***
	(0.003)	(0.002)	(0.003)	(0.002)	(0.007)	(0.004)	(0.004)	(0.002)	(0.005)	(0.002)
$\beta_1 + \beta_2$ ($\nu_{jt} < 0$)	0.026***	0.015***	0.024***	0.014***	0.032**	0.016***	0.035***	0.025***	0.029***	0.027***
	(0.006)	(0.002)	(0.004)	(0.003)	(0.014)	(0.003)	(0.006)	(0.002)	(0.005)	(0.002)
R^2	0.009	0.008	0.009	0.008	0.010	0.008	0.011	0.009	0.011	0.008
Obs. (Millions)	3.1	11.2	5.2	9.1	1.3	7.9	2.1	13.2	2.1	13.1
Equality Tests (p-values)										
$H_0 : \beta_1^R = \beta_1^{NR}$	0.131		0.474		0.181		0.053		0.735	
$H_0 : (\beta_1 + \beta_2)^R = (\beta_1 + \beta_2)^{NR}$	0.073		0.022		0.235		0.108		0.812	

Table C.2: Robustness: recession dating and firm shock measures. Estimates of equation (2) with asymmetric effects, by recession (R) and non-recession (NR) years, under alternative specifications. In columns 1–2 a year is recessionary if the Economic Tendency Indicator (ETI) was much weaker than normal in at least one quarter, and in columns 3–4 if it was weaker or much weaker than normal in at least two consecutive quarters. Columns 5–6 use a balanced panel of firms. Columns 7–8 measure firm shocks as log differences of productivity. Columns 9–10 estimate firm shocks from equation (1) without firm fixed effects. Standard errors clustered at the firm level in parentheses; $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Equality tests report p-values for the R–NR difference from a pooled specification interacting the regressors with the business cycle state.

Cobb–Douglas productivity measures. The baseline analysis measures firm productivity as log value added per worker. As a further check, I follow the literature on firm productivity estimation (see Syverson, 2011, for a review) and recover productivity as the residual $\tilde{z}_{jst}$ of a Cobb–Douglas production function,

$$\log Y_{jst} = a^N_{st} \log N_{jst} + a^K_{st} \log K_{jst} + \tilde{z}_{jst}, \tag{C.1}$$

where Y_{jst} is the value added of firm j in sector s at time t, N_{jst} and K_{jst} are labour and capital inputs, and a^N_{st} and a^K_{st} are factor shares. Implementing this approach requires a measure of each input and a method to estimate the factor shares. As labour input I use the firm's wage bill. For capital, I consider two alternatives: tangible capital, which like the wage bill is taken from the balance sheet data, and the perpetual inventory measure described in Appendix B. As for the factor shares, I estimate them in three ways. First, with the cost-shares method: the labour share a^N_{st} is the ratio of the total wage bill to total value added in each 2-digit sector s and year t, and the capital share is $a^K_{st} = 1 - a^N_{st}$.[46] Second, by OLS estimation of equation (C.1) within each 2-digit sector, retaining sectors with at least 20 firms in at least one sample year and a positive estimated capital coefficient. Third, with the method of Olley and Pakes (1996) — implemented with the R package *prodest* (Rovigatti, 2017) — which addresses the simultaneity bias of OLS production function estimation and partially mitigates measurement error. Since the methodology requires log investment and the lagged wage bill, I restrict the sample accordingly, again retaining sectors with at least 20 firms in at least one sample year and a positive estimated capital coefficient.

Having obtained each productivity measure, I estimate equation (1), recover the shocks, and run the pass-through regressions. Columns 1–6 of Table C.3 report the estimates: Panel A uses tangible capital, Panel B the perpetual inventory measure. The coefficients are generally smaller than in the baseline, as expected since the Cobb–Douglas residual removes the contribution of capital from the productivity measure, but the asymmetry is preserved throughout, with the equality test rejecting at the 10 percent level in four of the six cases and coming close to it (p-value 0.134) in the other two. Tacking stock, neither the cyclical asymmetry in the pass-through of negative shocks nor the acyclicality of that of positive shocks is an artifact of the measure of firm productivity adopted in the baseline.

Controlling for sectoral demand. A concern with value-added-based productivity is that it may reflect demand rather than physical efficiency. While firm-level price data are not available to me, De Loecker (2011) shows that prices can be accounted for by combining the production

[46] I exclude sector-year cells with fewer than 100 firms and set a^N_{st} to one whenever the estimated value exceeds unity.

function with a demand system. Letting output be $Y_{jt} = P_{jt}Q_{jt}$ with

$$Q_{jt} = \exp\{\tilde{z}_{jt} + u_{jt}\} \cdot N_{jt}^{\alpha_n} K_{jt}^{\alpha_k},$$
$$Q_{jt} = \exp\{\xi_{jt}\} \cdot Q_{st} \left(\frac{P_{jt}}{P_{st}}\right)^{\eta_s},$$

where P_{st} is the average sectoral price, Q_{st} a sectoral demand shifter, $\tilde{z}_{jt}$ firm productivity, u_{jt} measurement error, ξ_{jt} an unobserved firm-specific demand shock, and η_s a sector-specific demand elasticity, solving for P_{jt}, substituting, and taking logs yields

$$\tilde{y}_{jt} := y_{jt} - p_{st} = \beta_n n_{jt} + \beta_k k_{jt} + \beta_s q_{st} + \tilde{z}^*_{jt} + \xi^*_{jt}, \tag{C.2}$$

where $\beta_h = \alpha_h \left(\frac{\eta_s+1}{\eta_s}\right)$ for $h \in \{n, k\}$, $\beta_s = \frac{1}{\eta_s}$, $\tilde{z}^*_{jt} := \tilde{z}_{jt} \left(\frac{\eta_s+1}{\eta_s}\right)$, and $\xi^*_{jt} := \xi_{jt}/\eta_s$. Estimating this equation and using these relationships, I recover productivity as $(\tilde{y}_{jt} - \beta_n n_{jt} - \beta_k k_{jt} - \beta_s q_{st}) \cdot \frac{\eta_s}{\eta_s+1}$. The procedure does not fully isolate physical productivity: the residual retains the firm-specific demand component ξ^*_{jt}. Since this component is idiosyncratic, however, it should not systematically correlate with the business cycle, and is therefore unlikely to generate the spurious cyclical variation that motivates the exercise. Following De Loecker (2011), sectoral demand q_{st} is the weighted average of log sectoral value added, deflated with 2-digit sectoral prices, with weights given by each firm's share of sectoral sales. I estimate equation (C.2) by OLS for each 2-digit sector, with both capital measures, restricting to sectors with at least 20 firms in at least one year and positive estimated coefficients on capital and sectoral value added. Because the PPI is used to deflate value added and to construct sectoral demand, and PPI series are available throughout the sample only for manufacturing, the exercise is restricted to that sector (Table A.1 reports the baseline restricted to manufacturing for comparison). Columns 7–10 of Table C.3 report the estimates: as in the previous paragraph the coefficients are smaller than the manufacturing baseline, but the cyclical pass-through patterns are preserved.

Dependent variable: $\log \Delta w_{ijt}$, log earnings change										
	No sectoral demand control						Controlling for sectoral demand (Manufacturing only)			
	Cost shares		OLS		Olley-Pakes		OLS		Olley-Pakes	
	R	NR	R	NR	R	NR	R	NR	R	NR
Panel A: Tangible Capital										
β_1 ($v_{jt} \geq 0$)	0.012^{***}	0.014^{***}	0.012^{**}	0.021^{***}	0.014^{**}	0.021^{***}	0.021^{***}	0.018^{***}	0.021^{***}	0.018^{***}
	(0.004)	(0.002)	(0.006)	(0.002)	(0.006)	(0.002)	(0.007)	(0.003)	(0.007)	(0.004)
$\beta_1 + \beta_2$ ($v_{jt} < 0$)	0.009^{***}	0.003^{***}	0.023^{**}	0.004^{*}	0.023^{**}	0.004^{*}	0.021^{***}	0.005	0.021^{***}	0.005
	(0.003)	(0.001)	(0.010)	(0.003)	(0.010)	(0.002)	(0.007)	(0.003)	(0.008)	(0.004)
R^2	0.010	0.008	0.009	0.008	0.009	0.008	0.011	0.008	0.011	0.008
Obs. (Millions)	1.9	11.8	2.0	12.0	2.0	11.9	0.5	2.8	0.5	2.8
Equality Tests (p-values)										
$H_0 : \beta_1^R = \beta_1^{NR}$	0.767		0.195		0.245		0.641		0.613	
$H_0 : (\beta_1 + \beta_2)^R = (\beta_1 + \beta_2)^{NR}$	0.134		0.066		0.055		0.041		0.040	
Panel B: Perpetual Inventory Capital										
β_1 ($v_{jt} \geq 0$)	0.022^{***}	0.025^{***}	0.011	0.019^{***}	0.015^{**}	0.019^{***}	0.025^{***}	0.016^{***}	0.022^{***}	0.017^{***}
	(0.005)	(0.003)	(0.007)	(0.003)	(0.006)	(0.003)	(0.008)	(0.004)	(0.007)	(0.004)
$\beta_1 + \beta_2$ ($v_{jt} < 0$)	0.021^{***}	0.007^{***}	0.025^{**}	0.006^{**}	0.021^{**}	0.007^{***}	0.019^{**}	0.005	0.022^{***}	0.005
	(0.005)	(0.002)	(0.011)	(0.003)	(0.009)	(0.003)	(0.009)	(0.004)	(0.008)	(0.003)
R^2	0.010	0.008	0.010	0.008	0.009	0.008	0.012	0.008	0.012	0.008
Obs. (Millions)	1.8	10.7	1.8	10.7	1.7	10.5	0.4	2.4	0.5	2.6
Equality Tests (p-values)										
$H_0 : \beta_1^R = \beta_1^{NR}$	0.620		0.237		0.580		0.268		0.528	
$H_0 : (\beta_1 + \beta_2)^R = (\beta_1 + \beta_2)^{NR}$	0.008		0.094		0.134		0.106		0.038	

Table C.3: Robustness: alternative productivity measures. Estimates of equation (2) with asymmetric effects, by recession (R) and non-recession (NR) years, with firm productivity measured as the residual of a Cobb–Douglas production function and firm shocks then obtained from equation (1). Panel A uses tangible capital and Panel B uses capital constructed with the perpetual inventory method. Factor shares are estimated from cost shares at the 2-digit sector level (columns 1–2), by OLS within 2-digit sectors (columns 3–4), or following Olley and Pakes (1996) (columns 5–6). Columns 7–10 replicate the last two specifications controlling for sectoral demand following De Loecker (2011). Since sectoral PPI data are available throughout the sample only for manufacturing, they use manufacturing firms only; Appendix Table A.1 reports the baseline restricted to manufacturing for comparison. Standard errors clustered at the firm level in parentheses; $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Equality tests report p-values for the R–NR difference from a pooled specification interacting the regressors with the business cycle state.

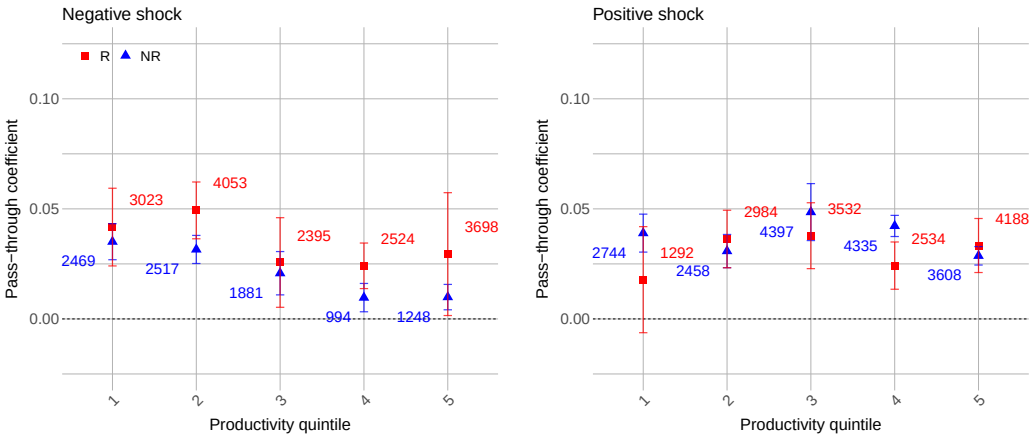
D Heterogeneity

This appendix complements the heterogeneity analysis of Section 2.3.2 by discussing the remaining dimensions: firm productivity, sector, and firm size (Panel A of Figure D.6), and age and education (Panel B). Each panel reports the estimates of equation (2) by subgroup for negative (left) and positive (right) shocks, in recessions and non-recessions. Panel B also reports the positive-shock estimates by tenure and earnings, whose negative-shock counterparts are in Figure 5.

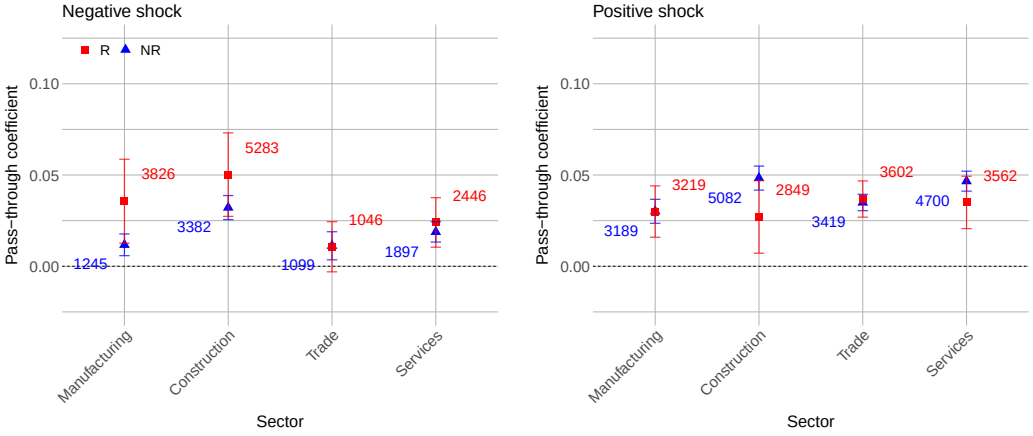
Starting with firm characteristics, the pass-through of negative shocks declines with lagged productivity in both states, while the gap in point estimates is roughly constant across quintiles (panel A.(a)): the cyclical erosion of insulation operates broadly across the productivity distribution rather than being concentrated in a specific part of it. By sector (panel A.(b)), the cyclical asymmetry is most pronounced in manufacturing and construction, followed by services and trade. Pass-through is monotonically declining in firm size at all points of the cycle (panel A.(c)), while the cyclical asymmetry is most pronounced at large firms. Turning to worker characteristics, the pass-through of adverse shocks declines with age in both states, but the gap in the point estimates is roughly constant across age groups (panel B.(a)). Likewise, a gap of similar size is present also in all education groups, and no gradient emerges across the different levels (panel B.(b)). For positive shocks, the pass-through shows no systematic gap between business cycle states in any dimension. Across groups it is broadly flat, with a decline in firm size and in age and a slight increase in education. Bearing in mind that the sample includes only stayers, and that selection into staying may matter more for some groups than for others, the cyclical asymmetry for adverse shocks is present in the point estimates across essentially all subgroups and does not seem to be driven by any particular dimension, apart from the concentration among workers with longer tenure and higher earnings discussed in the main text.

Panel A: Firm characteristics

(a) Productivity



(b) Sector



(c) Employment size

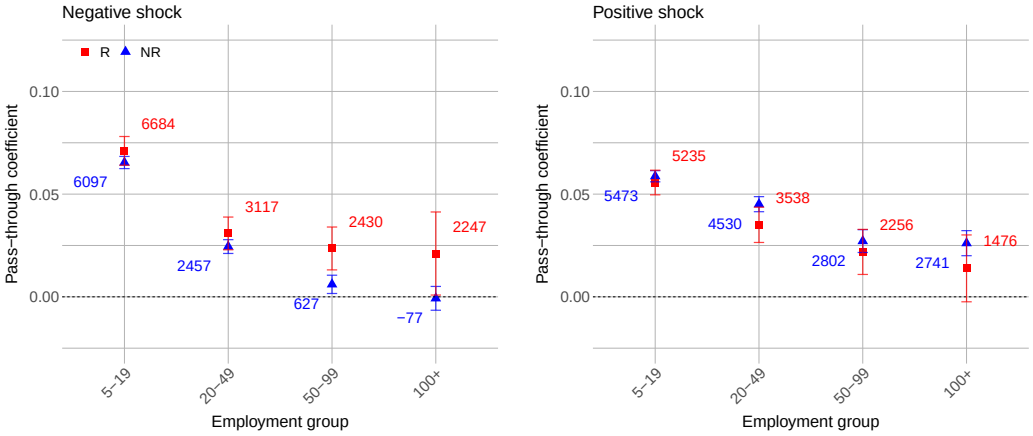
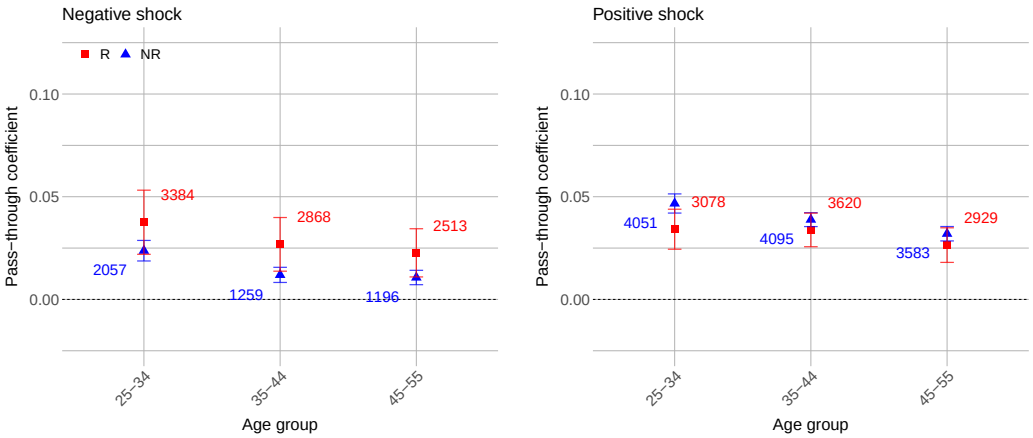


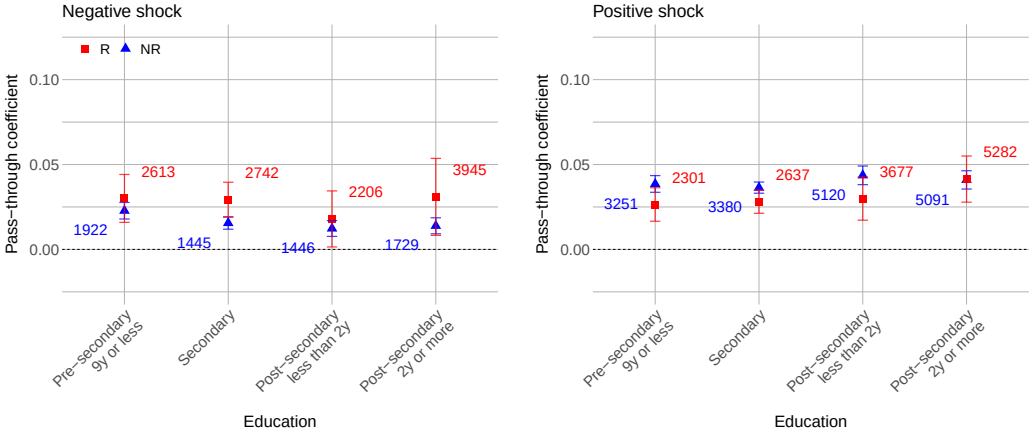
Figure D.6: Pass-through heterogeneity. Pass-through coefficients for negative and positive firm shocks by firm and worker characteristics, in recessions and non-recessions. Panel A, firm characteristics: (a) lagged productivity quintiles, (b) sector (manufacturing, construction, trade, services), (c) employment size (5–19, 20–49, 50–99, 100+). Panel B (next page), worker characteristics: (a) lagged age (25–34, 35–44, 45–55), (b) highest educational attainment, (c) lagged tenure, (d) lagged earnings quintiles. For tenure and earnings, only positive shocks are reported, as the negative-shock results are in Figure 5. Vertical bars denote 90% confidence intervals based on standard errors clustered at the firm level. Monetary values (2022 SEK) are computed as the coefficient times the standard deviation of the firm shock times average earnings in each group.

Panel B: Worker characteristics

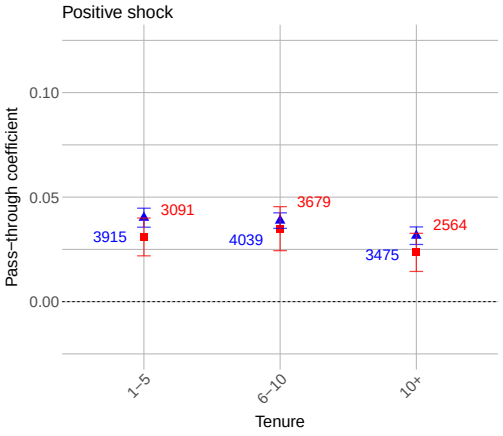
(a) Age



(b) Education



(c) Tenure, positive shock



(d) Earnings, positive shock

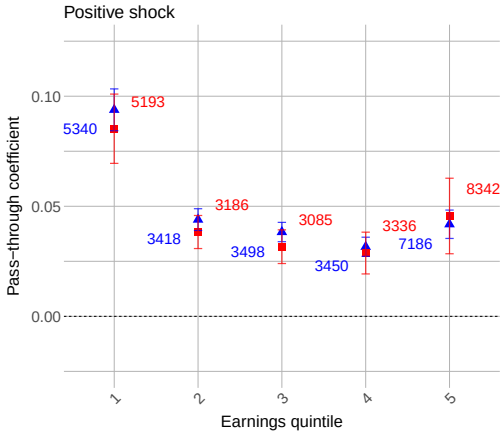


Figure D.6: (continued) See previous page for full caption.

E Numerical solution

E.1 Discretization and grids construction

The AR(1) process governing idiosyncratic firm productivity z_{jt} is discretized using the Rouwenhorst (1995) method with N_z points. Solving the model also requires a grid for promised utility and one for wages. I construct both using an uniformly spaced grid with N_x points. The extrema for the former are $u(b)/(1-\beta) - \varepsilon_x$ and $u(f(\overline{y},\overline{z}))/(1-\beta) + \varepsilon_x$, where $\overline{y}$ and $\overline{z}$ are the maximum values of aggregate and firm-specific productivity. The extrema for the latter are $w_{\min}^{\text{frac}} \cdot b$ and $w_{\max}^{\text{frac}} \cdot f(\overline{y},\overline{z})$. The grid for wages implicitly defines also a grid for $\mu = 1/u'(w)$.

In the numerical implementation, I set $N_z = 35$, $N_x = 501$, $w_{\min}^{\text{frac}} = 0.01$ (0.5 for the simpler model without on-the-job search), $w_{\max}^{\text{frac}} = 1$, and $\varepsilon_x = 1$. To increase accuracy when building some interpolation objects, I also construct a finer exponentially spaced grid with $N_{x,\text{large}} = 1001$ points and spacing parameter $\text{spacing}_{x,\text{large}} = 0.7$ to concentrate points near the upper end. To detect sign changes in the first-order condition I use a grid with 200 points with uniform spacing.

E.2 Solving the optimization problem

By the block recursivity result, the distribution of agents need not be included as a state variable. Summarizing the exogenous productivity states with $s = (y, z)$, the problem can be written as:

$$J(V,s) = \max_{w,\{\hat{V}(\hat{s})\}_{\forall \hat{s}}} f(y,z) - w + \beta(1-\delta)\mathbb{E}_{\hat{s}}\left[\left(1-\lambda_e \tilde{p}\left(\hat{V}(\hat{s}),\hat{y}\right)\right) J\left(\hat{V}(\hat{s}),\hat{s}\right)\right]$$

subject to:

$$V = u(w) + \beta\mathbb{E}_{\hat{s}}\left[\delta U(\hat{y}) + (1-\delta)\left(\hat{V}(\hat{s}) + \lambda_e R\left(\hat{V}(\hat{s}),\hat{y}\right)\right)\right] \tag{E.1}$$

The first-order conditions with respect to the current wage and next-period promised utility (for each state $\hat{s}$) are, respectively,

$$u'(w) = 1/\mu, \tag{E.2}$$

$$\frac{\lambda_e \tilde{p}'\left(\hat{V}(\hat{s}),\hat{y}\right)}{1-\lambda_e \tilde{p}\left(\hat{V}(\hat{s}),\hat{y}\right)} J\left(\hat{V}(\hat{s}),\hat{s}\right) = J'\left(\hat{V}(\hat{s}),\hat{s}\right) + \mu. \tag{E.3}$$

Combining them and using the envelope condition $-\mu = J'(V,s)$:

$$\frac{\lambda_e \tilde{p}'\left(\hat{V}(\hat{s}),\hat{y}\right)}{1-\lambda_e \tilde{p}\left(\hat{V}(\hat{s}),\hat{y}\right)} J\left(\hat{V}(\hat{s}),\hat{s}\right) = \frac{1}{u'(w)} - \frac{1}{u'(w(\hat{V}(\hat{s}),\hat{s}))} = \mu - \mu(\hat{V}(\hat{s}),\hat{s}) \tag{E.4}$$

I solve the model using the endogenous gridpoint method (Carroll, 2006). The fixed grid is the wage grid previously defined — which, recall, implicitly defines a grid for negative of the derivative of the firm value, μ — while the endogenous grid is over current promised utility V. Thus, every point in the grid for wages has a corresponding promised utility value. Let $J^{(i)}$ and

$V^{(i)}$ denote the current guesses for the firm value and current promised utility, respectively. The solution proceeds in the following steps:

1. Determine market tightness $\theta^{(i)}$ and the job finding probability $p^{(i)}$.
2. Compute the worker's search decision $m^{(i)}$ and the implied probability of leaving $\tilde{p}^{(i)}$.
3. Solve for the value of unemployment $U^{(i)}$.
4. Recover next-period promised utilities $\hat{V}^{(i)}$.
5. Compute updated current promised utilities $V^{(i+1)}$.
6. Determine the firm value implied by the new policies, $J^{(i+1)}$.
7. If the distance between successive policy and value functions falls below a predetermined threshold, stop; otherwise, return to step 1 with the updated guesses.

Each step is described in detail below. I first solve a simpler version of the model without on-the-job search or quits, then use the resulting solution as the initial guess for the full problem. In the simpler case, the initial guesses ($i = 0$) set current and future promised utility to $u(w)/(1-\beta)$, and the firm value to $\frac{f(\tilde{y},z)-w}{1-\beta(1-\delta)}$, where $\tilde{y}$ is the stationary value implied by the Markov chain governing aggregate productivity.

Market tightness. Let $\tilde{x}(y)$ denote the value of promised utility such that $k = J\left(\tilde{x}(y), y, z_0\right)$. Then, equation (9) pins down market tightness in submarket x:

$$\theta(x,y) = \begin{cases} q^{-1}\left(k/J\left(x,y,z_0\right)\right) & \text{if} \quad x \leq \tilde{x}(y) \\ 0 & \text{else.} \end{cases}$$

Solving the above for every point on the grid for $V^{(i)}$ and every aggregate productivity value y yields $\theta^{(i)}$, from which the matching function immediately delivers $p^{(i)}$. I locate $\tilde{x}(y)$, the promise $V^0(s)$ at which J crosses zero in each state, and the multiplier $\bar{\mu}(s)$ associated with $\tilde{x}$ by root-finding on linear interpolants of the relevant objects; the last two determine, respectively, the search interval and the case distinction in the firm's problem below. Since p is steep near $\tilde{x}(y)$, I evaluate it on the finer grid, interpolate it with the shape-preserving spline of Schumaker (1983), and fit a power function $a_0 + a_1(\tilde{x} - x)^{a_2}$ to obtain a smooth derivative for the worker's search problem.

Worker's search. Letting $f(x,V,y) = p\left(\theta\left(x,y\right)\right)\left(x - V\right)$ and following the same steps as in Menzio and Shi (2010), the optimal search policy can be summarized as:

$$m(V,y) = \begin{cases} \arg\max_{x \in X} f(x,V,y) & \text{if} \quad V < \tilde{x}(y) \\ V & \text{else.} \end{cases}$$

Since in $x \in (V, \tilde{x}(y))$, $f(x,V,y)$ is strictly concave (Menzio and Shi, 2010; Shi, 2009), if the solution is internal the first order condition pins it down:

$$p\left(\theta\left(x,y\right)\right) + p'\left(\theta\left(x,y\right)\right)\left(x - V\right) = 0.$$

I evaluate it with p from the spline and p' from the derivative of the fitted power function. The spline is used in levels because it passes through the computed values exactly, whereas the power function is not guaranteed to respect the bounds of a probability. The latter is an approximation for p', but a smooth one, which is necessary to avoid oscillations when solving the first-order condition.[47] I compute $m^{(i)}$ and $\tilde{p}^{(i)}(V, y) = p(\theta(m^{(i)}(V, y), y))$ on the fine grid and, as for p, interpolate $\tilde{p}^{(i)}$ with the Schumaker spline and fit a power function to obtain $\tilde{p}'^{(i)}$ for the firm's problem; $m^{(i)}$ is interpolated linearly.

Unemployment value. Next, I solve for the value of unemployment by iterating on the fixed point of equation (5). Starting from the initial guess $u(b)/(1-\beta) - \varepsilon_x$, at each iteration I evaluate the search return for every aggregate state with the worker's search decision policies constructed in the previous step and update the guess as specified by equation (5). The iteration terminates when the maximum absolute difference in the unemployment value across aggregate states between two consecutive iterates falls below the tolerance level.

Next-period promised utilities. For each point μ_i on the grid and each future state $\hat{s} = (\hat{y}, \hat{z})$, I solve for the next period promised utility $\hat{V}^{(i)}(\hat{s})$ using the first-order condition:

$$-\lambda_e \tilde{p}'^{(i)}\left(\hat{V}, \hat{y}\right) J^{(i)}\left(\hat{V}, \hat{s}\right) + \left(1 - \lambda_e \tilde{p}^{(i)}\left(\hat{V}, \hat{y}\right)\right)\left(\mu_i - \mu^{(i)}\left(\hat{V}, \hat{s}\right)\right) = 0.$$

The structure of the solution depends on whether μ_i is below or above $\bar{\mu}(\hat{s})$, the value of μ corresponding to $\tilde{x}(\hat{y})$.

***Case A*: $\mu_i < \bar{\mu}(\hat{s})$.** The solution lies in the friction region $\hat{V} < \tilde{x}(\hat{y})$: above it the first-order condition reduces to $\mu_i = \mu(\hat{V}, \hat{s})$, which cannot hold since μ is increasing and $\mu(\hat{V}, \hat{s}) \geq \bar{\mu}(\hat{s}) > \mu_i$ there. Within the friction region, let V^* satisfy $\mu(V^*, \hat{s}) = \mu_i$. Since $\tilde{p}' < 0$ and $(1 - \lambda_e \tilde{p}) > 0$, the sign of J determines the sign of $(\mu_i - \hat{\mu})$ at the solution: when $J > 0$ the firm optimally raises $\hat{V}$ above V^* to reduce poaching, and when $J < 0$ it lowers $\hat{V}$ below V^* to increase poaching. The solution therefore lies between V^0 and V^*, with the interval capped at $\tilde{x}(\hat{y}) - \epsilon$, where ϵ is a small tolerance that keeps evaluations strictly inside the friction region. I evaluate the condition on a grid over this interval, take every crossing from above as a candidate local maximum, located precisely by a bracketed root-finder, and add the lower end of the interval when the condition is still negative there, or the upper end when it is still positive there, since the objective then peaks at the boundary. Among the candidates I select the one maximizing the constraint-substituted objective $(1 - \lambda_e \tilde{p})J + \mu_i(\hat{V} + \lambda_e R)$, whose derivative is the first-order condition.

***Case B*: $\mu_i \geq \bar{\mu}(\hat{s})$.** In this case there is always a solution in the region $\hat{V} \geq \tilde{x}(\hat{y})$ where frictions are inactive, obtained by solving $\mu(\hat{V}, \hat{s}) = \mu_i$; since μ is increasing in $\hat{V}$ there, this

[47] I use the same approach also when using $\tilde{p}$ to solve the first-order condition for future promised utilities below.

root is a local maximum. Further local maxima may exist in the friction region, but only where $J < 0$, that is, in $[V^0, \tilde{x}(\hat{y}) - \epsilon]$ when $V^0 < \tilde{x}(\hat{y})$. I scan this interval as in Case A, collect all crossings from above and any outward-pointing corner as candidates, and select, among these and the frictionless root, the one maximizing the same objective.

Current promised utilities. Next, for each $s = (z, y)$, I use the promise keeping constraint (7) to compute the updated V for each point in the grid for μ. That is, use the policies and values computed at the previous step to compute:

$$V^{(i+1)}(s) = u(w_i) + \beta \mathbb{E}_{\hat{s}} \left[\delta U^{(i)}(\hat{y}) + (1-\delta) \left(\hat{V}^{(i)}(\hat{s}) + \lambda_e R^{(i)} \left(\hat{V}^{(i)}(\hat{s}), \hat{y} \right) \right) \right].$$

This is the endogenous grid step: we are finding the V that must have generated $\hat{V}$ computed at the previous point as optimal policies. Note that we use w_i since, as previous explained, the grid for μ is 1-to-1 with the grid for w.

Firm value. The last step within each iteration is to update the firm value at each point V_i implied by the μ grid and each current state $s = (y, z)$ using the Bellman equation and the policies computed above:

$$J^{(i+1)}(V_i, s) = f(a, y, z) - w_i + \beta(1-\delta) \mathbb{E}_{\hat{s}} \left[\left(1 - \lambda_e \tilde{p}^{(i)} \left(\hat{V}^{(i)}(\hat{s}), \hat{y} \right) \right) J^{(i)} \left(\hat{V}^{(i)}(\hat{s}), \hat{s} \right) \right].$$

Convergence and updating. After computing the new policies and values, I check convergence by comparing the updated objects to those from the previous iteration using the maximum norm. Convergence is declared when the maximum absolute difference across the firm value J, the promised utility V, the next period promised utility policy $\hat{V}$, and the unemployment value U falls below a tolerance level (10^{-5}). If convergence has not been reached, I replace the current guesses with the newly computed values.

E.3 Simulating the model

This appendix describes how the simulated panel is constructed and how the empirical procedure of Section 2 is applied to it. I simulate 10,000 workers over 1,000 quarters, discarding the first 200 as burn-in. The aggregate state is drawn once per period from the two-state chain of Section 4 and is common to all workers, who start unemployed. Workers then move between unemployment and matches according to the model's timing, with new matches starting at the entry productivity z_0 and at the promise selected by the worker's search policy, and continuing matches carrying the promised utility given by the policy $\hat{V}$. Each match receives an identifier so that employment spells can be tracked.

The model is quarterly and the data annual, so the simulated series are then aggregated to the annual frequency. A year is classified as recessionary if the economy is in the recessionary state in at least three of its four quarters. Mirroring the six-month rule applied to the registers,

a worker is assigned in each year to the match in which she spends the largest number of quarters, provided this is at least two, with ties broken in favour of the match paying the higher total wage; a worker who spends fewer than two quarters in any single match is classified as non-employed for that year. Annual earnings are the average quarterly wage paid by the assigned match, multiplied by four, so that the earnings of workers who are with the firm for part of the year are annualized as in the data, and annual firm productivity is constructed in the same way from the quarterly realizations $f(y, z)$ of that match; non-employed workers receive $4b$. Annual transitions are defined by comparing match identifiers across consecutive years: a worker matched in $t-1$ and in t to different matches is recorded as a job-to-job mover, one matched in $t-1$ but not in t as an employment-to-unemployment transition, and the reverse as a transition out of unemployment. Tenure accumulates as long as the identifier is unchanged. For job-to-job movers, the productivity of the origin firm in the transition year is reconstructed by locating the quarter in which the worker leaves that match and annualizing the idiosyncratic draw realized in it, which is the object used in Figure 7.

On the annualized panel I follow the empirical procedure of Section 2. Idiosyncratic firm shocks are the residuals of equation (1), estimated with match fixed effects in place of firm fixed effects, since each firm employs one worker, and with year fixed effects in place of sector-by-year effects, since the model economy has a single sector. As in the data, only matches with at least five usable observations are retained. The pass-through regression (2) is then estimated on stayers, with the subset of controls that have a model counterpart, trimming earnings changes and firm shocks at the 0.5th and 99.5th percentiles within each period and clustering standard errors at the match level.

F Estimation

The goal of the minimum distance estimation routine is to find a vector of parameters Φ that minimizes the objective function defined by:

$$\sqrt{\frac{d(\Phi)'\Omega d(\Phi)}{\text{trace}(\Omega)}}$$

where $d(\Phi)$ is a vector containing deviations of model moments from their targets in the data and Ω is a weighting matrix. I rescale the weighted moment deviations by the trace of the weighting matrix and take the square root of the resulting quantity. In this way, when Ω is diagonal, the resulting value can be interpreted as the weighted quadratic mean of the moment deviations.

The estimation procedure includes a global and a local stage and it is similar to the TikTak algorithm developed by Arnoud et al. (2019). In the global stage, I generate a Sobol sequence of length N_{glo} over the parameter space and compute the objective function for each vector of parameters in the sequence. In the local stage, I use the best N_{loc} points from the global

stage — i.e., the points achieving the lowest objective function values in the global stage — as starting guesses for running a local search with the Nelder-Mead algorithm. The global stage allows to identify the most promising regions of the parameter space to find the optimum, thus minimizing the resources allocated for the computationally intensive local search algorithm.[48] The code implementing the estimation routine just described can be found in a dedicated Julia package (Azzalini and Rácz, 2024).

I apply the above procedure to estimate the firm-specific productivity process via indirect inference. In the structural model, log firm productivity is $\tilde{z}_{jt} = y_t + z_{jt}$, where y_t is the aggregate component and $z_{jt} = \rho_z z_{jt-1} + \varepsilon_{jt}$, with $\varepsilon_{jt} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_\varepsilon^2)$, is a firm-specific AR(1) process. In this exercise, the aggregate component takes the two values implied by the observed ratio of productivity in recessions to normal times, 0.968, rather than the calibrated one, which stands in for the broader forces that compress match surplus in downturns (Section 4) and therefore implies a larger drop than the one observed in aggregate productivity.[49] The two parameters $(\rho_z, \sigma_\varepsilon)$ are estimated by simulating a panel of 10,000 firms over 500 quarters. N_{glo} and N_{loc} are chosen to be 10,000 and 100, respectively. The last 60 quarters reproduce the aggregate states observed in the data, with the recessionary state assigned to 2008Q3–2009Q3 and the non-recessionary state elsewhere, while the preceding quarters serve as a burn-in to avoid dependence of the results on the initial conditions (the initial z is set to zero for all firms). Since the model operates at quarterly frequency, annual productivity is obtained by summing $e^{\tilde{z}_{jt}}$ over the four quarters of each year and taking the logarithm of the resulting sum. The regression specified in equation (1) is then estimated on the simulated panel over the data period, and $(\rho_z, \sigma_\varepsilon)$ are chosen to match the unconditional variance of $\tilde{z}$ and the variance of the residual ν to their data counterparts. The procedure is repeated for 10 independently simulated economies, each with its own draw of shocks, and the resulting moments are averaged across economies.

[48] Note that while the global stage searches in a bounded region of the parameter space decided by the researcher — the one for which the Sobol sequence is created — the local search step is not bounded and can potentially deliver any point.

[49] The ratio is computed from real quarterly GDP, log-linearly detrended, as the average over 2008Q3–2009Q3 divided by the average over the remaining quarters. With the normalization $\mathbb{E}[e^y] = 1$ it implies $e^{y_{\text{NR}}} = 1.003$ and $e^{y_{\text{R}}} = 0.971$.